

Financial Engineering Master Thesis Report

14.08.2026

News-Based Systematic Strategies

Evidence from an Accessible Provider with Point-in-Time Language Models

Student:	Rocco Pio Lorenzo VENTRUTO
Company:	Taranis
FE supervisor:	Professor Semyon MALAMUD
Company supervisor:	Julien LEEGENHOEK

Abstract

Large language models extract return-predictive signals from financial news, but the evidence rests almost entirely on premium newswires few researchers can access. This thesis asks whether accessible news carries a predictive signal, and whether the results established there extend to it. We replicate the point-in-time methodology of [Kelly et al. \(2026\)](#), from chronologically consistent embeddings to monthly news portfolios, with the pure news design of [Didisheim et al. \(2026\)](#), which purges the news embeddings of the firm characteristics that predict them. We do so on 1.6 million single-stock Benzinga articles distributed by Massive, over two point-in-time universes, the Nasdaq-100 and a broad universe of more than two thousand firms. Three results emerge. First, a genuine signal exists on the broad universe, with significant gross alphas against the market throughout, but the pure news anomaly of [Didisheim et al. \(2026\)](#) does not extend to our corpus: performance does not strengthen as the embeddings are purged of ever more firm characteristics. Second, we quantify the *Premium Gap*, the price paid, in Sharpe ratio terms, for replacing the premium newswires of our reference literature with an accessible provider: positive throughout, and largest at the deepest purges. Finally, we find that the headline carries the signal and the body dilutes it: under headlines alone, alphas are significant against factor models as demanding as the 13 theme portfolios of [Jensen et al. \(2023\)](#), and the market alpha stays significant net of realistic transaction costs.

Contents

1	Introduction	6
2	Related Work	9
3	News Provider Selection	16
4	Data	18
4.1	Sources	18
4.2	Preprocessing	19
4.3	Sample and Universe Definition	21
5	Feature Construction	22
6	The Estimation Sample	25
7	News Predictability from Firm Characteristics	27
8	Portfolio Construction	29
9	Results	31
9.1	The Nasdaq-100	31
9.2	The Broad Universe	33
9.3	Residualization Depth	34
9.4	Factor Exposures Along the Residualization Ladder	36
9.5	Firm Size and Pure News Portfolio Performance	40
9.6	The Premium Gap	42
9.7	Pooled Sorts under Squared-Error Selection	45
9.8	Alternative Shrinkage Selection Rules	47
9.9	News Input and Pure News Performance	49
9.10	Transaction Costs	52
10	Robustness	54
10.1	Article-to-Month Aggregation	54
10.2	Point-in-Time Models Comparison	55
10.3	The Ensemble Volatility-Scaling Convention	56

10.4 Performance by Subperiod	57
10.5 Rolling Scheme and Lookahead Bias	58
11 Conclusion	61
References	65
Appendices	69
A Padding and the Reproducibility of News Embeddings	69
B Residualization Mapping	70
C Additional Tables	72
Generative AI Disclaimer	75

List of Figures

1	The Editorial Shift: News Volume, Automation, and Coverage	26
2	The Common Component of the Embeddings	27
3	The Common Component Under Rolling Checkpoints	28
4	Predicting News Embeddings from Firm Characteristics	29
5	News Portfolio Performance on the Nasdaq-100	32
6	News Portfolio Performance on the Broad Universe	34
7	Pure News Sharpe Ratios and the Number of Characteristics	35
8	Pure News Sharpe Ratios by Size Group	40
9	Pure News Sharpe Ratios and Cross-Sectional Breadth	42
10	Pure News Sharpe Ratios Across Selection Rules	47
11	Pure News Sharpe Ratios Across News Inputs	50
12	Pure News Sharpe Ratios Across Vintage Checkpoints	59

List of Tables

1	Article Counts Through the Data Pipeline	22
2	Pure News Portfolio Exposures to the Fama–French Six Factors	37
3	Pure News Portfolio Exposures to the JKP Theme Portfolios	38
4	The Premium Gap: The Price of Accessible News	43
5	Pure News Portfolio Sorts Under MSE Selection	46
6	Pure News Portfolio Sorts Under Sharpe Selection	48
7	Pure News Portfolio Alphas Under the Baseline and Headline-Only Inputs . . .	51
8	Pure News Net-of-Cost Performance	53
9	Pure News Sharpe Ratios Across Embedding Aggregation Conventions	55
10	Pure News Sharpe Ratios Across Point-in-Time Models	56
11	Pure News Sharpe Ratios Across Volatility-Scaling Conventions	57
12	Pure News Sharpe Ratios by Subperiod	58
B.1	Residualization Specifications and Their JKP Characteristics	71
C.1	Pure News Portfolio Sorts on the MSRR Weights	72
C.2	Pure News Portfolio Sorts Under Rank-IC Selection	73
C.3	Pure News Portfolio Sorts Under Tile Spread Selection	74

1 Introduction

Large language models offer a new way to read the news. An article is no longer a bag of words, its terms counted against a predefined dictionary (Tetlock, 2007; Tetlock et al., 2008) or a list of words selected by returns (Ke et al., 2019). They allow a nuanced and contextual understanding of the text, condensed into a single numerical representation: an embedding.

Chen et al. (2022), henceforth CKX, bring this representation to asset pricing and show that portfolios built on news embeddings outperform their word-based predecessors, opening a line of work that has since grown around this design. Didisheim et al. (2026), henceforth DKPT, take it one step further: much of what news reports is already predictable from prevailing firm characteristics, so that a raw embedding mixes genuinely new information with content the market could have anticipated. Purging that predictable part leaves what they call “pure news”, a residual that predicts returns far more strongly than the raw embedding it comes from, and the more characteristics are purged, the stronger the signal becomes. Purged of the full set of characteristics, it delivers what they name the pure news anomaly. Both results, however, rest on a commercial newswire that few researchers can access, and in this they mirror the field.

The financial NLP literature relies almost entirely on a handful of premium institutional providers, such as the Dow Jones Newswires (Ke et al., 2019; He et al., 2025; Kelly et al., 2026; Engelberg et al., 2026), Reuters (Chen et al., 2022; Glasserman and Lin, 2023; Didisheim et al., 2026), and the Wall Street Journal (Tetlock, 2007; Manela and Moreira, 2017; Bybee et al., 2024). These are sources with an established track record in published work, whose data quality can reasonably be taken as given.

Accessible news sources, by contrast, those within the reach of an individual researcher, barely appear in the literature at all (Dong et al., 2024; Gu et al., 2024). This silence is itself uninformative: it may reflect genuine limitations, or merely the fact that researchers with institutional access to premium feeds have had little incentive to test the alternatives. Either way, whether the findings established on premium newswires carry over to accessible sources remains untested. The answer matters beyond budget-constrained researchers: well-funded teams, too, have reason to validate a proof of concept on accessible data before committing to a premium subscription.

Testing this question faces an obstacle of its own. Recent work has documented the capacity of language models to memorize and recover information from their training corpus (Sarkar and

Vafa, 2024; Lopez-Lira et al., 2025; Engelberg et al., 2026). This capacity casts doubt on any predictive result built on such models: part of the measured performance may reflect what the model memorized from future data rather than what was known at the time of the decision, a contamination known as lookahead bias or training leakage. Point-in-time, or chronologically consistent, language models remove the problem at its source by providing a sequence of dated checkpoints, each trained only on text available on or before a given date: ChronoBERT and ChronoGPT (He et al., 2025), DatedGPT (Yan et al., 2026), and the PIT series of Kelly et al. (2026), henceforth **KMSX**. Among these, **KMSX** train at the largest scale, and the embeddings of their PIT-4B models yield stronger portfolios than those of ChronoGPT. Adopting the pure-news concept of **DKPT**, their news application works at the monthly frequency and residualizes embeddings on firm characteristics.

This thesis takes that toolkit to accessible news. From **KMSX** it takes the point-in-time extractor and the pipeline that follows it, from embeddings to portfolios, and from **DKPT**, the pure news design. The questions are whether accessible news predicts returns, and whether purging news of its predictable content strengthens the signal. The source is a Benzinga newswire feed distributed by Massive, 1.6 million single-stock articles spanning 2009 to 2025, evaluated over two point-in-time universes: the Nasdaq-100 and a broad universe averaging more than two thousand firms a month.

We show that the answer is split. News does predict returns on an accessible feed, and best when read from the headline alone; but the signal does not strengthen as more characteristics are purged from it. Additionally, replacing the premium newswires with an accessible provider carries a price, which we quantify.

Contributions. *First*, we show that the pure news anomaly documented by **DKPT** does not extend to our corpus, despite firm characteristics explaining a share of our news embeddings similar to what they report. Making the signal increasingly orthogonal to known characteristics buys no performance. The one large performance gain is the cross-sectional demean, whereas the full JKP residualization, the configuration that carries the literature’s headline claims, is generally the weakest.

Second, we quantify the *Premium Gap*: the price, in Sharpe ratio terms, of replacing premium newswires with an accessible provider, under methodological configurations closely aligned with our reference papers **KMSX** and **DKPT**. We show that the gap is positive wherever a comparison

benchmark exists, and concentrates at the deepest residualization levels, where it is corroborated by two distinct premium newswires, Dow Jones and Reuters.

Third, we establish that a genuine signal nonetheless exists in accessible news, with portfolios earning gross alphas significant against the market factor throughout, and against the Fama–French six factors in most configurations. It is, however, a broad-universe phenomenon: on the Nasdaq-100, even the best configuration is at most marginally significant against the QQQ benchmark, a weakness that persists across configurations and estimators.

Fourth, we show that the headline carries the signal and the body of the articles dilutes it on our broad universe: restricting the input to headlines strengthens the alphas against every factor model considered, and achieves significance against the strictest of them, the 13 theme portfolios of [Jensen et al. \(2023\)](#), where the full text earns none. Net of realistic transaction costs, this configuration is the only one that retains a market alpha significant at the 5% level. It is also an order of magnitude cheaper to embed, a saving directly relevant to the audience an accessible provider might serve.

Fifth, we contribute to the reproducibility of our reference methodology itself, [KMSX](#). In the course of the replication we documented several inconsistencies: the paper and its released code disagreed on the monthly aggregation convention of the embeddings, the scaling of the MSRR effective shrinkage penalty, and the shrinkage grid. More consequentially, the released extraction code left-padded the tokenized news sequences, a choice under which the paper’s results did not replicate. We reported these findings to the authors, who confirmed them.

2 Related Work

Text as data. A large literature treats text as data in economics and finance ([Gentzkow et al., 2019](#)), and return-relevant information has been documented across corpora: corporate filings ([Loughran and McDonald, 2011](#)), earnings calls ([Price et al., 2012](#)), analyst reports ([Lv, 2024](#)), and social media ([Bollen et al., 2011](#)). This thesis concerns news, which the literature has approached first through dictionary-based and word-count methods and, more recently, through neural models of text, today dominated by transformers.

Word-count methods. Dictionary-based and word-count methods provided the first evidence that news text predicts returns. [Tetlock \(2007\)](#) links a daily measure of media pessimism, built from word counts in a Wall Street Journal column, to market returns: high pessimism predicts downward price pressure followed by a reversion to fundamentals. At the firm level, [Tetlock et al. \(2008\)](#) show that the fraction of negative words in firm-specific news stories forecasts low firm earnings, with stock prices briefly underreacting to the information these words embed. Subsequent work documents the temporal structure of this predictability: investors react even to stale, previously published information, with price reactions that partially reverse ([Tetlock, 2011](#)). The evidence extends beyond the United States and beyond short horizons: [Calomiris and Mamaysky \(2019\)](#) document news-driven predictability of country index returns in 51 developed and emerging economies, with stronger predictive power at the one-year horizon than at one month.

Within this word-count paradigm, a separate line of work estimates the word weights directly against market data. [Ke et al. \(2019\)](#) learn both the word list and the word weights from stock returns on the Dow Jones Newswires: words are retained when they appear disproportionately in articles accompanied by positive or by negative returns, and their sentiment weights are then estimated using returns as training labels. The resulting sentiment scores predict the next-day returns of the covered firms. In the same spirit, [Manela and Moreira \(2017\)](#) estimate a weight for each n-gram of Wall Street Journal front-page articles by regressing option-implied volatility on their frequencies. The resulting index predicts market returns at horizons of six months to two years.

Transformer-based methods. Transformer-based models changed the representation itself: rather than counting words, they encode each article in context, preserving word order and word interactions. These models have given rise to a new strand of the news-based literature.

A first line of work keeps the sentiment scheme of the earlier literature: language models further pre-trained on financial news and fine-tuned on sentences annotated for sentiment, such as FinBERT (Araci, 2019), improve on both word-count and earlier neural methods for financial sentiment analysis. Lopez-Lira and Tang (2023) remove the adaptation step altogether: they prompt GPT-4 to assess whether a news headline is good or bad for the stock price of the firm it concerns. The resulting scores capture the direction of the initial market response and predict one to two days of subsequent return drift, especially for small stocks and negative news. Across the language models they examine, forecasting ability generally increases with model size.

A second line of work takes its supervision from the market itself: rather than labels reflecting human or model judgment, the returns that follow each article drive the estimation. Kirtac and Germano (2024) fine-tune OPT and BERT on up/down labels given by the sign of the three-day excess return following each article’s publication: trading strategies built on the fine-tuned models’ scores outperform those based on the off-the-shelf FinBERT and on the Loughran-McDonald dictionary, with OPT performing best. Going one step further, Guo and Hauptmann (2024) remove the binary labelling step altogether: using LoRA, they fine-tune encoder-only and decoder-only LLMs (DeBERTa, Mistral, Llama) to map a stock’s recent newsflow directly to its one-month forward return; in experiments spanning North American, European, and emerging-market universes, portfolios built on these forecasts generally outperform those based on conventional sentiment scores.

Frozen embeddings. Fine-tuning is only one way of coupling the news with returns. The same supervision can instead be applied outside the model: the pre-trained LLM is kept frozen and serves as a feature extractor, with all task-specific learning left to a downstream prediction model. The appeal of this division of labour is economic: whereas fine-tuning requires dedicated GPU resources for every model and every experiment, pre-training has been carried out once by the model’s developers, and the resulting representation can be reused by researchers who have neither the expertise nor the computing budget to train or adapt a large model.

CKX introduce this design in asset pricing: theirs is the first paper to model expected returns with news embeddings extracted from large language models. Their implementation follows two steps. Each article of a Refinitiv corpus of single-stock news is converted into an embedding by averaging the token-level representations produced by a given model, for six language models of successive generations (BERT, RoBERTa, the LLaMA family, and OpenAI’s embedding model).

A pooled panel ridge regression then maps the embeddings to next-day open-to-open returns, and daily long-short quintile portfolios are formed on the predictions.

The embedding-based portfolios outperform those built on word-based representations, including Word2Vec, the Loughran-McDonald dictionary, and the SESTM model of [Ke et al. \(2019\)](#), with the most advanced models performing best. Formal tests of predictive accuracy confirm the gains over the word-count benchmarks, while the most advanced models are statistically indistinguishable from one another. The signal is not subsumed by short-term reversal or by firm characteristics and survives realistic transaction costs. This predictability reflects underreaction: prices keep moving in the direction of the news for several days after publication, a drift that is strongest among small stocks and fades almost immediately for large ones, consistent with limited investor attention and limits to arbitrage.

The advantage of embeddings over word-based methods also holds outside the United States, in a pooled sample spanning fifteen equity markets and news written in twelve languages. Both the paper and its method have become the reference point of a growing literature: subsequent work adopts the two-step design and varies its open choices.

[DKPT](#) take it one step further: much of what news reports is already predictable from prevailing firm characteristics, so that a raw embedding mixes genuinely new information with content the market could have anticipated. Purging that predictable part leaves what they call “pure news”, a residual that predicts returns far more strongly than the raw embedding it comes from, and the more characteristics are purged, the stronger the signal becomes. Purged of the full set of characteristics, it delivers what they name the pure news anomaly.

The anomaly exceeds every other documented in the cross-section: portfolios formed by penalized maximum Sharpe ratio regression (MSRR) ([Kelly and Xiu, 2023](#)) on pure news earn an annualized Sharpe ratio of 3.1 over their 1996–2022 sample, against 1.4 for the best factor of the JKP anomaly universe. It is also unusually long-lived, predicting returns up to 18 months after the news arrives. Reading the embeddings through sparse autoencoders, they trace the anomaly to investor underreaction to negative-tone and quantitative topics, and to overreaction to high-attention and ambiguous ones. The finding survives an extensive robustness program. It is robust to the extractor, with a monotonic relationship between model size and performance, the Sharpe ratio rising from 1.5 with BERT to 4.1 with the largest Llama model; and to the news source, reaching 3.7 on the Dow Jones Newswires and 2.3 even on the third-party news their quality filters exclude.

Accessible news sources. Accessible sources appear only at the margins of the news-based literature. [Gu et al. \(2024\)](#) come closest to our data configuration with a Benzinga feed of 843,062 headlines on Nasdaq-100 stocks over 2009–2020. A pre-trained FinBERT scores each headline into positive, neutral, and negative probabilities, averaged by day; these sentiment features, alongside the past eight closing prices, feed an LSTM that forecasts the next day’s closing price. Adding the news signal improves the mean absolute percentage error of the price-only benchmark from 7.2% to 4.5%.

[Dong et al. \(2024\)](#) assemble FNSPID, an open corpus of 15.7 million news records time-aligned with prices for 4,775 S&P 500 firms over 1999–2023, built by scraping the Nasdaq.com news pages and merging the result with earlier open collections (Bloomberg and Reuters dumps, Benzinga, and the Russian-language Lenta.Ru corpus). The collection process leaves visible gaps: timestamps are dates without a publication time, only 37.5% of the records carry a stock symbol, and 24.8% include the article body. The evaluation is a forecasting benchmark: news enters as a ChatGPT sentiment score, averaged by day, which deep learning models add to past prices and volumes to forecast short-horizon closing prices. The news signal improves accuracy only modestly, mostly confined to the transformer-based model.

Neither exercise builds portfolios, adjusts for risk, or enforces point-in-time discipline. There is little evidence on whether accessible news translates into portfolio performance at all: Sharpe ratios, alphas against a benchmark, or returns that survive transaction costs.

Lookahead bias. Financial LLM research has grown faster than its evaluation practice: surveying 164 papers published in main conferences between 2023 and 2025, [Kong et al. \(2026\)](#) find that only 26.8% explicitly discuss lookahead bias, the leakage that arises when a system uses information that was unavailable at decision time, whether memorized in model weights or supplied by external context. We focus on the first, parametric channel, and review the evidence in two steps: the capacity of language models to memorize and recover pre-cutoff information and the mitigations available.

[Lopez-Lira et al. \(2025\)](#) query LLMs on a broad financial corpus (macroeconomic series, market indices, CRSP stocks, WSJ headlines, earnings calls, RavenPack news) and find that the models recall financial facts predating their knowledge cutoffs, in some cases to the exact value; the memorized information extends to embeddings. [Sarkar and Vafa \(2024\)](#) reach the same conclusion through a different route: a language model prompted with the candidates’

biographies picks the winner of close U.S. House races, outcomes assumed unpredictable ex ante, 70 to 80 percent of the time.

A first family of defenses acts on the text supplied to the model, in increasingly aggressive forms. Prompt-level instructions fail first. [Sarkar and Vafa \(2024\)](#) prompt a language model with pre-pandemic earnings calls and the instruction to consider only information available at the time of the call, and find that generated risk forecasts still mention COVID-19, a sequence that did not exist at the time. [Lopez-Lira et al. \(2025\)](#) instead impose a knowledge cutoff earlier than the model's true one and ask factual questions about the period in between: the model answers with near-perfect accuracy when the cutoff is stated in the system prompt, and even when the model appears to respect the cutoff, there is no way to verify that the memorized knowledge is not being used.

Removing identifiers fares little better. With the firm's name censored, a model re-identifies the subject firm from the remaining context in roughly 70 percent of cases, on earnings calls ([Sarkar and Vafa, 2024](#)) as on news articles ([Engelberg et al., 2026](#)). Widening the mask helps only marginally: censoring the firm's products alongside its name lowers identification to 61 percent on earnings calls ([Sarkar and Vafa, 2024](#)), and the systematic version of this idea, masking up to 20 firm-related terms drawn from the Google Knowledge Graph ([Glasserman and Lin, 2023](#)), still leaves 47.3 percent of firms identifiable on news articles ([Engelberg et al., 2026](#)). Time is no easier to hide than identity: the year of a date-censored earnings call is recovered with a correlation of 0.79 ([Sarkar and Vafa, 2024](#)), and under name-only masking the publication date of a news article is recovered to within a week 92.5 percent of the time ([Engelberg et al., 2026](#)).

Entity neutering closes the ladder. Rather than enumerating what to hide, [Engelberg et al. \(2026\)](#) let the model anonymize the text itself: one instance masks identifying terms and rewrites the article, an independent instance then attempts to name the firm, and the text goes through another round until identification fails. On a sample of over 500,000 Dow Jones news articles, firm identification falls to 0.11 percent, the rate of random guessing, while the sentiment extracted from the neutered text agrees with that of the raw text more than 95 percent of the time. The defense has costs: roughly \$1,500 per 100,000 articles in API calls, and part of the semantic content is lost in the process. More fundamentally, identification at chance levels remains only a lower bound: failure to recover the firm does not prove that no memorized information is being used ([Lopez-Lira et al., 2025](#)).

Point-in-time language models. These defenses all share the same premise: the model has memorized the future, and the text must be stripped of whatever lets it access that knowledge. The evidence above shows how far this can be pushed and where it ends: partial masking fails, and the procedure that succeeds is costly, loses part of the information, and can never prove that memorized knowledge is truly out of play. The memorization itself, however, originates in the pretraining corpus, and a second family of solutions attacks the problem at its source: point-in-time, or chronologically consistent, language models. These models come as a sequence of dated checkpoints, each trained only on text available at its own date, so that every analysis period can be paired with a model whose training ended before it.

[He et al. \(2025\)](#) provide an implementation with ChronoBERT (149M parameters) and ChronoGPT (1.5B), an encoder built on a modernized BERT architecture and a decoder built on a modified nanoGPT, released as annual checkpoints from 1999 to 2024, each trained exclusively on text published before its declared cutoff. The initial checkpoints are pretrained on a corpus of seven billion tokens of open-source English text published before 2000, repeated over several epochs, the number depending on the model; each subsequent checkpoint then continues training on the following year’s text alone, drawn from an incremental corpus of 65 billion tokens through 2024. On general language benchmarks, even the earliest ChronoBERT checkpoint reaches a GLUE score of 84.71, above the original BERT (84.52) and well above FinBERT (76.41), within two points of Llama 3.1 (86.31), a model more than fifty times its size.

In their news application, on the Dow Jones Newswires, firm-level news is aggregated into embeddings following [CKX](#) and mapped to next-day return forecasts. The point-in-time element is the extractor itself: under a rolling scheme, each year’s articles are embedded by the checkpoint of the previous year, so that the model advances with the corpus while always predating it. Portfolios sorted on the resulting forecasts significantly outperform those built on BERT, FinBERT, and StoriesLM, another chronologically trained model ([Sarkar, 2024](#)), and are statistically indistinguishable from those of Llama 3.1. Chronological consistency thus comes, in their setup, at little visible cost.

Scale is the question [KMSX](#) take up, and their models separate its two margins. PIT-1.5B holds ChronoGPT’s parameter count and scales the chronological data alone, to 170 billion FineWeb tokens. PIT-4B scales the model itself, to 4 billion parameters trained on one trillion tokens. Both margins pay: zero-shot HellaSwag accuracy moves from 44% for ChronoGPT to 58% for

PIT-1.5B and to 72% for PIT-4B, within a few points of Gemma-3-4B (77%) and LLaMA-7B (76%), open-weight models trained without temporal restriction. The advantage is not specific to HellaSwag: across the seven zero-shot common-sense reasoning and language comprehension benchmarks they report, PIT-4B outperforms every other point-in-time model on every available task. A chronologically constrained model thus improves with data and parameters like any other language model: what remains of the gap to its unconstrained counterparts, the authors argue, is a scale deficit rather than an inherent limitation of the point-in-time paradigm.

To improve downstream usability, they additionally instruction-tune PIT-4B with LoRA, yielding a third variant, PIT-4B-FT; the instruction data are filtered to remove references to events beyond each cutoff, so that temporal consistency is preserved through post-training as well. On IFEval, PIT-4B-FT outperforms both ChronoGPT-Instruct and Qwen1.5-1.8B, the latter trained without temporal restriction.

Their news application returns to the Dow Jones Newswires, at the monthly frequency. Articles are embedded under the same rolling scheme and aggregated into firm-month embeddings. These are residualized on firm characteristics into pure-news portfolios, following [DKPT](#), then combined by penalized MSRR and ensembled over a grid of shrinkage parameters. Under this scheme, PIT-4B-FT outperforms PIT-4B, and both outperform the ChronoGPT variants, base and instruct. This thesis builds on this pipeline, with PIT-4B-FT as its embedding model.

How large is the bias these designs eliminate? In the embeddings channel, the evidence points to a minor concern. [KMSX](#) and [DKPT](#) reach the same verdict from a single comparison each, the rolling scheme, free of lookahead bias by construction, against a model whose training window covers the entire sample period: enforcing the point-in-time constraint costs little or nothing. In [DKPT](#) the two Sharpe ratios are indistinguishable. In [KMSX](#) the point-in-time models in fact outperform their full-sample counterparts (1.53 against 1.18 for PIT-4B-FT).

[He et al. \(2025\)](#) provide the most systematic evidence. Rather than a single pair, they evaluate the entire series of vintage checkpoints: if lookahead bias were material, the final vintage, with the most comprehensive knowledge, would be the top performer across all periods. Instead, they find an envelope: for ChronoGPT, the rolling scheme outperforms every vintage, even those with later knowledge cutoffs. They suggest that temporal alignment drives this dominance: a rolling scheme interprets each article with the language of its time, whereas a later vintage might read old news with meanings those words only took on later.

3 News Provider Selection

No ground truth. Choosing a news source is not like choosing a price vendor. For price data, vendor comparison rests on standardized criteria, and the underlying quantity has a canonical ground truth: a closing price is a single value, up to minor adjustment conventions, verifiable against the exchange's record, so that errors in a low-cost source are detectable and correctable. News admits no ground truth. There is no reference set of articles about a given stock on a given day, only a partial view from each provider, differing in coverage, depth, and bias. The provider does not merely transmit the data, it shapes what reaches the researcher: through the content itself (what is covered, how well, and whether by a specialized reporter or an automated template) and through the metadata it attaches, from ticker tagging to categorization and timestamping. The choice is therefore a first-order methodological decision rather than a logistical one, and one for which the literature offers no established basis of comparison. The criteria applied here are set out explicitly.

Provider landscape. One distinction organizes them. Under *first-party distribution*, the provider delivers content it has written itself, controlling the entire chain from production to delivery, including the metadata attached to each article, such as publication timestamps and entity tagging. Under *aggregation*, the provider ingests and redistributes content produced by others; its metadata may come from the original producer, from the aggregator's own tagging and timestamping, or from a combination of the two. This is not, in itself, a defect, but each additional step away from the source can reduce transparency, notably the ability to trace an article back to who published it, when, and as originally written. These are distribution methods, not mutually exclusive categories: a single provider may do both. Benzinga, for instance, writes most of what it serves in-house while also carrying press releases and a small share of relayed items. How content reaches a researcher does not by itself settle whether a source is suitable. Suitability rests on several considerations, and they are not equally weighty: most, such as noise, coverage gaps, or uneven tagging, concern how much a source degrades the signal and involve trade-offs rather than thresholds. One class of considerations, however, is decisive.

Point-in-time integrity. At any date, the archive should reflect what was on the record then; without this, no result built on the data is methodologically reliable. Three things can drift after the fact, each pulling the record away from what was on file at the time. The first is the tag attaching an article to a security. Every accessible provider we examined tags its articles with a

ticker symbol, the short exchange code, such as AAPL or MSFT, under which a stock trades. That symbol is not a stable key: it is reassigned, retired, and reused over time. The company a ticker identifies today is therefore not necessarily the one it identified when an article was written. The second is the text itself: an article may be revised after publication, and if the archive silently stores the amended version under the original date, the timestamp no longer describes the text it labels. The third is the archive, which may not preserve the past at all: entries tied to companies since delisted or acquired may be pruned, leaving a record that reflects today's surviving universe rather than the one in place then.

To illustrate, take Facebook, which became Meta and changed its ticker from FB to META in 2022. A source that retags retroactively returns its pre-2022 articles under META, as if the company had always traded under that symbol. The same change breaks the query in the other direction: FB was later reassigned to an unrelated ETF, so a query on FB now returns articles about that fund. Companies that disappear leave no trace at all: for firms delisted through acquisition, such as Celgene or Activision, the news history is neither retrievable under the retired symbol nor attached to the acquirer's.¹ None of these failures announces itself: an empty query looks like sparse coverage, and a mistagged or recycled symbol returns well-formed articles about the wrong company.

Requirements. From these failure modes we derive three point-in-time requirements a source must meet. First, tags must be frozen at publication: an article must remain attributable to the company that carried its ticker when it was written. Second, the text and its date must agree: the archive must either preserve the article as first published or date the version it serves. Third, the archive must be complete: articles tied to companies since delisted, acquired, or renamed must remain in the record. Three further conditions are practical rather than methodological: the full text of each article, title and body, must be available; the archive must reach far enough back for a backtest to span several market regimes; and the source must be verifiable, through a testable free tier or, failing that, documented guarantees of point-in-time behavior.

Provider selection. For several of the providers we examined, the requirements could not be assessed at all: no free tier to test them, documentation silent on how corporate events are handled, and support inquiries left unanswered. A guarantee that cannot be obtained is, for our purposes, equivalent to a failure. Across the sources we tested, historical depth was a recurring

¹These failure modes were observed in our own tests of the [Alpha Vantage](#) news API in April 2026 and are undocumented by the provider.

limitation: premium archives reach back before 2000, while most accessible providers offer little or nothing before 2010. The screening leaves two candidates: the news API of [EODHD](#) and a Benzinga feed redistributed by [Massive](#). Both serve the full article text, title and body, and both meet the point-in-time requirements. The former, however, is a pure aggregator: 85% of our free-tier sample resolved to Yahoo Finance links, itself an aggregate. We proceed with Massive. Its market news is mostly written in-house by a single newsroom, so the feed reflects one editorial policy, evolving over time but uniform at any given date and documentable, rather than a cross-section of policies inherited from whichever publisher an aggregator collects. Its ticker tags are frozen at publication, revisions are dated rather than silent, and its archive reaches back to 2009 and retains the history of companies since delisted or acquired, which we verified on the corpus and the provider confirmed on inquiry.

4 Data

The study period runs from August 2009 to November 2025 in formation months: the lower bound is set by the start of news coverage, the upper bound by the last monthly formation for which the one-month-ahead return target is available. The corresponding returns are realized through December 2025.

4.1 Sources

News. Our news data consist of financial news articles published by Benzinga, redistributed by [Massive](#). Each article carries a provider-assigned identifier (`benzinga_id`), a headline, the body text where present, a list of ticker tags, the author, two timestamps, and auxiliary metadata we do not use. The timestamps are `published`, the original publication time, and `last_updated`, the time of the article’s most recent revision (hereafter the *revision time*); the text served is the version as of that time.

Index membership and identifiers. Nasdaq-100 membership is obtained via Wharton Research Data Services (WRDS) from the Compustat historical index constituents file (`idxcst_his`), which records the start and end date of every membership spell, with firms identified by Compustat’s permanent company identifier (`gvkey`). Our unit of observation is the security rather than the firm: securities are identified by the permanent identifier (`permno`) of the Center for Research in Security Prices (CRSP), stable through ticker changes and corporate events. Two

linking tables complete the identifier infrastructure: the CRSP/Compustat Merged database (CCM, `ccmxfp_lnkhist`), which maps `gvkey` to `permno`, and CRSP’s historical security file (`stocknames_v2`), which records, for each `permno`, the tickers it carried and over which periods.

Returns and firm characteristics. Stock-month level data are obtained from the Global Factor Data (Jensen et al., 2023) via WRDS: the one-month-ahead excess return (in excess of the risk-free rate), the size-group classification (mega, large, small, micro, nano), and the 153 main firm-level characteristics.

Factors and benchmarks. The factors are the Fama–French five (Fama and French, 2015): the market excess return (Mkt-RF), size (SMB), value (HML), profitability (RMW), and investment (CMA), augmented with momentum (Mom). Their monthly returns and the one-month Treasury-bill risk-free rate are obtained from the Kenneth R. French Data Library. Additionally, monthly returns of the 13 theme portfolios of Jensen et al. (2023) are obtained from the Global Factor Data website. Finally, monthly total returns of the Invesco QQQ Trust (QQQ), an exchange-traded fund replicating the Nasdaq-100 index, are obtained from CRSP (`msf_v2`).

4.2 Preprocessing

The feed arrives in raw form and requires preprocessing before analysis. Retrieval covers August 2009 through November 2025. This yields 2,692,068 articles.

Format conversions. Timestamps are converted from UTC to US/Eastern, the timezone in which all trading-day logic is defined, and titles and bodies, delivered as HTML, are parsed to plain text (BeautifulSoup with the lxml parser). Four record-level irregularities are then cleaned: malformed records, publication–revision intervals, duplicate identifiers and ticker tags, and extreme lengths. Each count below is measured on the corpus entering the stage; the filters are then applied jointly, and as a few articles fall under more than one, the individual counts slightly exceed the total removed.

Malformed records. Articles whose headline is duplicated verbatim as the body (376; 0.014%) are short-form items in which the body adds nothing: we blank the body and retain the headline. Articles with neither headline nor body (11; <0.01%) carry no text and are dropped.

Publication–revision interval. We examine the gap between publication and revision times to detect irregularities. Articles whose revision time precedes publication (90; <0.01%) carry a

corrupted timestamp and are dropped. The distribution is otherwise concentrated near zero: 94% of articles are either never revised or revised within one minute of publication, so anchoring on the revision time coincides with the publication time for the overwhelming majority of the corpus. At the other end, a thin tail exceeds 14 days (6,758; 0.25%), dominated by 2009–2011 articles whose revision dates cluster heavily. Whatever these late updates were, anchoring on them would date potentially stale content as fresh news; we drop articles whose revision lag exceeds 14 days, a horizon beyond the regular editorial cycle.

Duplicate identifiers and ticker tags. Benzinga re-emits a small set of almost identical evergreen articles under the same article identifier, predominantly its “This Day in Market History” series republished on each anniversary. We retain the earliest emission per identifier, removing 518 articles (0.019%), after which the article identifier is a unique key. Separately, some articles repeat the same symbol within their ticker list (5,998; 0.22%), an ingestion artifact: we collapse each list to its distinct symbols, so that the ticker count reflects the firms an article actually references, which matters for the count-based filtering below.

Length filtering. Following [CKX](#), whose filter [DKPT](#) also adopt, we discard articles exceeding 100,000 characters (67; <0.01%). We impose no lower bound. The 100-character floor of [CKX](#) is not applied to short alerts, the headline-only breaking-news items a dedicated tag identifies in their corpus. No field in our feed sets such items apart. Manual inspection shows they arrive as ordinary articles whose body follows no systematic pattern: absent altogether, reduced to a source attribution such as “-Reuters”, or longer. Any length rule meant to isolate them would be arbitrary.

Single-ticker filtering. Of the 2,684,667 articles surviving the cleaning steps, most carry exactly one ticker tag (2,077,794; 77.4%). A further 162,394 (6.0%) carry none; these are predominantly non-equity content, such as macroeconomic releases, foreign exchange, or political coverage. Among the tagged articles, we retain only single-ticker items, as do [CKX](#), [KMSX](#), and [DKPT](#). The restriction isolates firm-level news: each retained article is unambiguously attributed to exactly one firm.

Firm resolution. Each remaining article is mapped from its ticker tag to a permno through the `stocknames_v2` file: the tag is attributed to the firm whose ticker-validity window covers the calendar day of the article’s revision time. Tags matching no listed security on that day are dropped; these are predominantly cryptocurrencies and currency pairs tagged as symbols.

Duplicate filtering. The last preprocessing concern is redundancy: re-emitted articles that carry no new information. CKX address it by discarding any article whose bag-of-words cosine similarity to another article from the prior five business days exceeds 0.8, with the stated aim of removing repeated coverage of the same events. This instrument does not transfer to our corpus: Benzinga’s templated series produce distinct events under near-identical boilerplate, so textual similarity signals templating rather than redundancy, and a similarity filter would discard genuinely distinct information². We therefore retain the CKX window but restrict removal to exact duplicates: articles of the same firm whose title and body are character-for-character identical to an earlier article within the window; the earliest emission is retained. The window itself is essential: templated text is deterministic, so identical articles can recur months apart while describing distinct events.³

4.3 Sample and Universe Definition

The analysis operates on monthly cross-sections of news, formed at the close of each month’s last trading day, hereafter the *rebalance date*. Each article is therefore assigned to a month, hereafter its *assigned month*: the calendar month of its revision time. An article is deferred to the following month when its news can no longer inform the rebalance, that is, when the revision time falls within the final two hours before the month’s last close⁴, after that close, or on a weekend or holiday ending the month. The two-hour margin is a conservative execution assumption, representing the time needed to process the news. Within its assigned month, each article is then placed on a day, hereafter its *assigned day*: the calendar day of its revision time or, for deferred articles, the first calendar day of that month, keeping day and month consistent. Finally, each article must map to a target at its assigned month, the firm’s excess return over the following month; Table 1 traces the article counts from the raw feed through this stage.

We consider two investable universes, both defined at each month’s rebalance date; the two universes are treated separately throughout. The broad universe comprises the US common

²Two of Benzinga’s automated options alerts on Amazon.com, issued five days apart, share near-identical boilerplate yet report opposite events: the first flags bullish sweep calls with the price up at \$3,615.65, the second bearish sweep puts with the price down at \$3,434.09. A similarity filter would conflate them.

³For instance, “Openheimer Maintains Outperform on Netflix, Maintains \$725 Price Target” appears in our corpus with byte-identical title and body on two dates roughly three months apart, two separate analyst notes whose templated text coincides because neither the rating nor the target changed in between. Removing the second would discard a real event.

⁴The cutoff accounts for early-close sessions, using the NASDAQ schedule from the `pandas_market_calendars` library (v5.3.2).

Table 1: Article Counts Through the Data Pipeline

This table reports the article count after each stage of the data pipeline described in the text.

Raw	Cleaning	Single-Ticker	Firm Resolution	Duplicate Filtering	Return Target
2,692,068	2,684,667	2,077,794	2,022,396	2,013,402	1,745,210

stocks belonging to the mega, large, small, or micro size groups; only the nano group is excluded. The Nasdaq-100 universe comprises the index members; membership spells, natively keyed by `gvkey`, are mapped to `permno` through the CCM link table.

An article is retained only if its firm belongs to the universe at the rebalance date of its assigned month. In the Nasdaq-100, the stock must additionally have been a member when the news was issued. A stock leaving the index mid-month thus has its entire exit month discarded; for a stock joining mid-month, only the articles issued from its entry onward are retained. The broad-universe sample contains 1,633,481 articles and the Nasdaq-100 sample 182,453; both are used in the subsequent steps.

5 Feature Construction

Embedding extraction. We transform the textual content of each news item into an embedding, its numerical representation. Embeddings are extracted with PIT-4B-FT, the instruction-tuned variant of the point-in-time language model PIT-4B of [KMSX](#) and the best performing extractor in their own news application. For each article, the input text is the title and the body joined by a period, or the title alone when the body is blank; the tokenized sequence is truncated to the model’s 2,048-token context window, which affects 0.17% of the corpus.

The defining property of this model is that it is released as a series of checkpoints, each trained only on data up to a given date. By selecting, for each article, a checkpoint that postdates none of the article’s own date, we ensure the feature extractor has been exposed only to information that existed when the article was written, so that the representation itself cannot be contaminated by future knowledge absorbed during pretraining.

An article dated in year y is embedded with the December $y-1$ checkpoint, an assignment we refer to as the *rolling-checkpoint scheme*: the extractor strictly predates the text and advances

with the corpus rather than remaining fixed.⁵ For revised articles we observe only the latest revision, which may retain portions written at publication; anchoring the checkpoint on the publication year is therefore the conservative choice, guaranteeing that the model’s training data predates every portion of the text, original or revised. Articles dated before 2014 precede the earliest checkpoint and are embedded with it; these months serve only as training history, and the out-of-sample window is constrained accordingly.

Articles are processed in batches: shorter sequences are right-padded to the batch maximum with a filler token. The embedding itself is the model’s final hidden state after RMS normalization, taken at the article’s last real token. This padding convention is not innocuous: Appendix A documents a batch-dependence issue in the released extraction code, which we identified and the authors confirmed and corrected. Formally, letting $P = 4,096$ denote the dimension of this representation, each article a is mapped to an embedding $e_a \in \mathbb{R}^P$.

Embeddings aggregation. For each universe, we aggregate the per-article embeddings of its retained sample into a monthly stock-level representation in two stages, using the assigned days and months of Section 4.3.

For each (stock, day) pair, the daily embedding is the equally weighted average of the embeddings of the sample’s articles of that stock with that assigned day:

$$\bar{e}_{d,s} = \frac{1}{|\mathcal{A}_{d,s}|} \sum_{a \in \mathcal{A}_{d,s}} e_a \in \mathbb{R}^P,$$

where $\mathcal{A}_{d,s}$ denotes the set of sample articles of stock s with assigned day d .

For each (stock, month) pair, the monthly embedding is the equally weighted average of these daily embeddings, taken over the days on which the stock has at least one article:

$$\tilde{e}_{t,s} = \frac{1}{|\mathcal{D}_{t,s}|} \sum_{d \in \mathcal{D}_{t,s}} \bar{e}_{d,s} \in \mathbb{R}^P,$$

⁵Checkpoints obtained from Hugging Face, repositories Diamegs/PIT-4B-FT- $\{\text{vintage}\}$, one per training vintage from December 2013 to December 2024; <https://huggingface.co/Diamegs>. The 2015 vintage is dated November rather than December, the underlying FineWeb corpus having no December 2015 snapshot.

where $\mathcal{D}_{t,s}$ is the set of days in month t on which stock s has at least one article.⁶ When $\mathcal{D}_{t,s} = \emptyset$, the monthly embedding $\tilde{e}_{t,s}$ is undefined and stock s is excluded from month t 's cross-section.

By construction, $\tilde{e}_{t,s}$ is point-in-time at the month boundary: deferred articles carry the following month as their assigned month (Section 4.3), so no article arriving within the final two hours before a rebalance, or after it, enters the embedding traded on it.

Residualization. The monthly embedding $\tilde{e}_{t,s}$ summarizes the textual content of the news concerning firm s during month t . This content, however, is not independent of what is already known about the firm: it may reflect characteristics whose predictive power is well established in the literature. A strategy built on the raw embedding would then risk deriving its performance from the mere replication of these known risk factors and anomalies rather than from the news-specific information itself. To isolate the news-specific signal, we follow DKPT and remove from each embedding the component explained by these documented predictors, retaining the orthogonal residual.

We process the 153 JKP characteristics of Section 4.1 following KMSX,⁷ within each investable universe (Section 4.3).⁸ First, we retain the 130 best-covered JKP characteristics over the universe's full sample, dropping the 23 with the highest share of missing observations; among these, we drop stock-months with more than 30% missing values. The retained characteristics are then rank-standardized within each month and rescaled to $[-0.5, 0.5]$, with remaining missing entries imputed to zero, the cross-sectional mean rank. We collect the characteristics entering a given residualization specification into $S_{t,s} \in \mathbb{R}^K$, the characteristics vector of firm s in month t , where K depends on the specification and reaches $K = 130$ when the full characteristic set is used.

For each month t , we run the cross-sectional regression⁹ of embeddings on characteristics across the firms of the universe with both a monthly embedding and a characteristics vector that month;

⁶Our implementation follows the released code of KMSX (<https://github.com/DjoFE2021/Point-In-Time-LLM>); the published equation (§4.3) uses instead $|\mathcal{D}_t|$ as the denominator, where $\mathcal{D}_t$ is the set of trading days in month t (a stock-independent denominator). The released code averages over the days with news for stock s ; the authors confirmed in personal correspondence that this is a typo in the published equation and that the code corresponds to the reported results.

⁷KMSX adopt the coverage screen of Didisheim et al. (2024), which selects 130 characteristics; DKPT use a closely related set of 132.

⁸JKP does not cover the entire Nasdaq-100 universe: ADRs, for example, have no JKP data and exit the panel at this stage.

⁹The regression is fit in closed form with a small ridge term ($\lambda_0 = 10^{-8}$) added to the Gram matrix $S_t' S_t$, a numerical safeguard against near-collinearity among the characteristics rather than a shrinkage of the signal. It is computed in double precision.

the same cross-section is used at every residualization specification:

$$\tilde{e}_{t,s} = \alpha_t + \beta_t S_{t,s} + \hat{e}_{t,s}, \quad \beta_t \in \mathbb{R}^{P \times K}, \quad \alpha_t \in \mathbb{R}^P.$$

The regression decomposes each *raw embedding* $\tilde{e}_{t,s}$ into a *predictable news* component $\hat{e}_{t,s} = \alpha_t + \beta_t S_{t,s}$, the part spanned by known characteristics, and an orthogonal *pure news* residual $\hat{e}_{t,s} \in \mathbb{R}^P$, the part that is not (terminology follows [DKPT](#)). These three objects are what we study. Pure news is defined relative to the conditioning set $S_{t,s}$: each residualization specification yields its own.

We residualize the embeddings against a ladder of characteristic sets of increasing richness: *Constant* (cross-sectional demeaning); *CAPM* adds the market beta; *FF3* adds market capitalization and book-to-market; *FF6* adds profitability, investment, and momentum; *JKP13* completes the thirteen JKP themes with one characteristic each; and *JKP* uses the full 130. The first five are nested; *JKP*, defined by coverage, is not in general a strict superset of *JKP13*. We introduce *JKP13* because the full *JKP* residualization is over-parameterized on the Nasdaq-100’s narrow cross-section, where the residual degenerates; *JKP13* keeps the regression well-posed while spanning all thirteen themes. Its composition is fixed ex ante, held constant across universes and runs; the selection rule and full mapping are described in [Appendix B](#).

6 The Estimation Sample

This section characterizes the estimation sample of [Section 4.3](#) restricted to the stock-months that pass the characteristics coverage screen of [Section 5](#). The screen removes 0.2% of the Nasdaq-100 articles and 3.9% of the broad-universe articles, leaving 182,055 and 1,569,216 respectively. [Figure 1](#) summarizes the resulting input over time.

The feed ramps up over its first two years, then both volume and coverage are stable through 2019. Since 2010 is a full year, the ramp is not an artifact of the partial 2009 window. From 2020 the feed undergoes an editorial shift. Volume rises sharply on both universes, and part of the rise is directly attributable to automated production: from that year the feed carries two disclosed automated authors, *Benzinga Insights* and *Benzinga Neuro*, whose stories are machine-generated and then reviewed by staff editors.

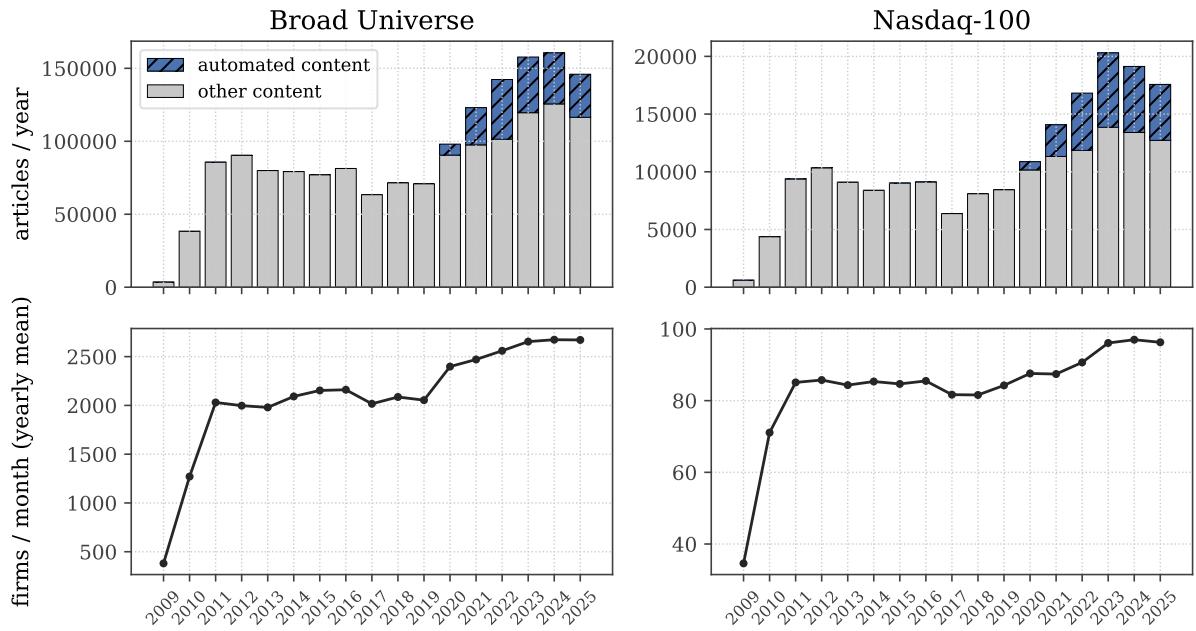


Figure 1: The Editorial Shift: News Volume, Automation, and Coverage

Note. This figure characterizes the estimation sample on both universes. The top row reports the number of articles per year, split between automated content, identified from the provider’s two disclosed automated authors, and all other content. The identified share is a lower bound. The bottom row reports the average number of firms per monthly cross-section. 2009 covers August–December and 2025 January–November.

They account for a large share of the volume from 2020 onward. The bottom row shows the counterpart in coverage: the broad cross-section widens from about 2,050 firms per month in 2019 to about 2,670 by 2024, the widening concentrated over the same period, and the Nasdaq-100 rises to nearly the full index.

On average over the window, the feed carries about 96,000 articles per year on the broad universe and 11,100 on the Nasdaq-100, over cross-sections averaging 2,155 and 85 firms per month; the average Nasdaq-100 member is thus covered about three times more densely.

7 News Predictability from Firm Characteristics

Before turning to portfolios, this section measures the share of the monthly embeddings' cross-sectional content that firm characteristics explain on our estimation sample (Section 6), the component the residualization ladder subsequently removes. Under rolling point-in-time checkpoints, the measurement itself requires care.

Figure 2 (top row) shows that the raw embeddings are dominated by a common component: the pooled uncentered adjusted R^2 is 97.6% at every specification on the Nasdaq-100 and between 96.3% and 96.6% on the broad universe. Almost all of this share is the work of the intercept alone.

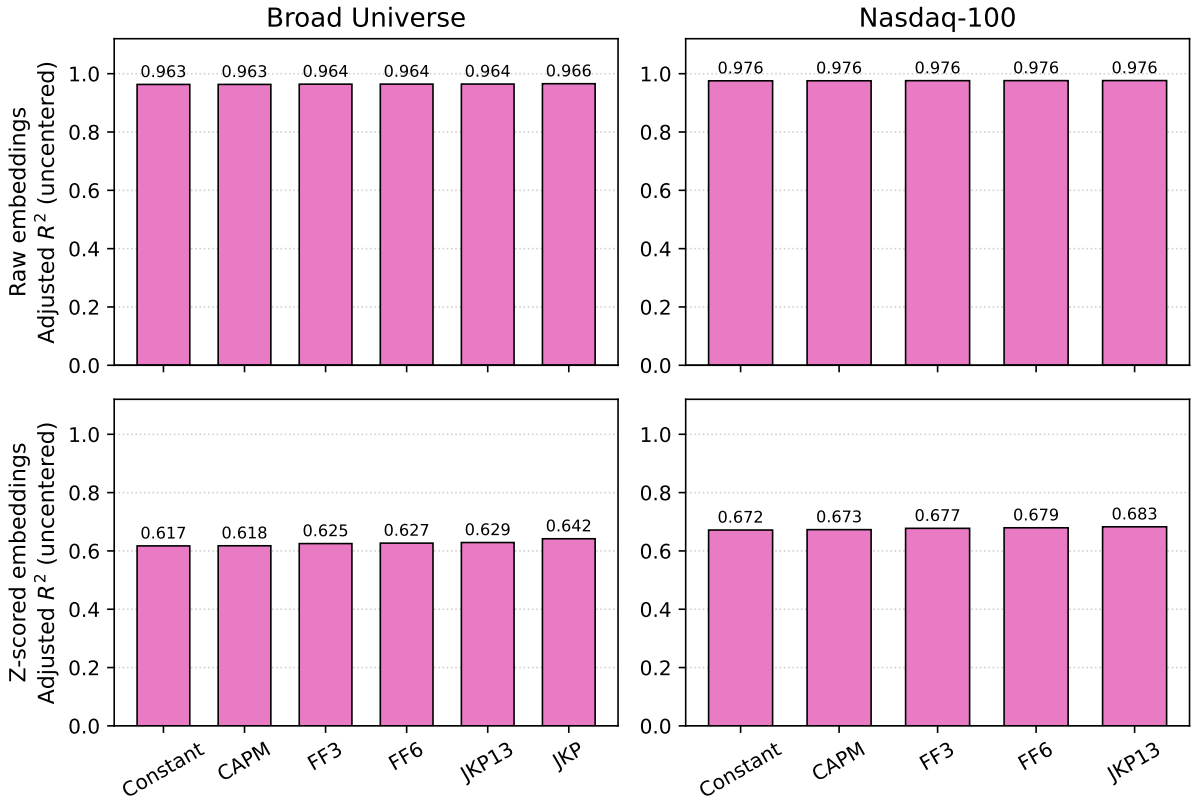


Figure 2: The Common Component of the Embeddings

Note. This figure shows the pooled adjusted uncentered R^2 from monthly cross-sectional regressions of the news embeddings on the regressors of each residualization specification, on both universes. The top row uses the raw embeddings, the bottom row the embeddings standardized with the expanding Z-score of [DKPT](#).

[DKPT](#) standardize each embedding coordinate with an expanding Z-score to counter the anisotropy of LLM embeddings, and note that the resulting coordinates have “nearly zero mean”, the premise that justifies their uncentered R^2 . On our corpus, the same transformation falls short of it (bottom row). It does absorb part of the common component, lowering the intercept share, but what

remains is far from nearly zero, and at every conditioning set almost the entire fit is the monthly intercept.

Figure 3 identifies the cause. The intercept share of the Z-scored embeddings jumps at each January checkpoint switch; when the same stock-month cells are rebuilt from embeddings extracted with a single fixed vintage, the staircase disappears and the share falls by an order of magnitude, to a flat floor of roughly 5–15%.

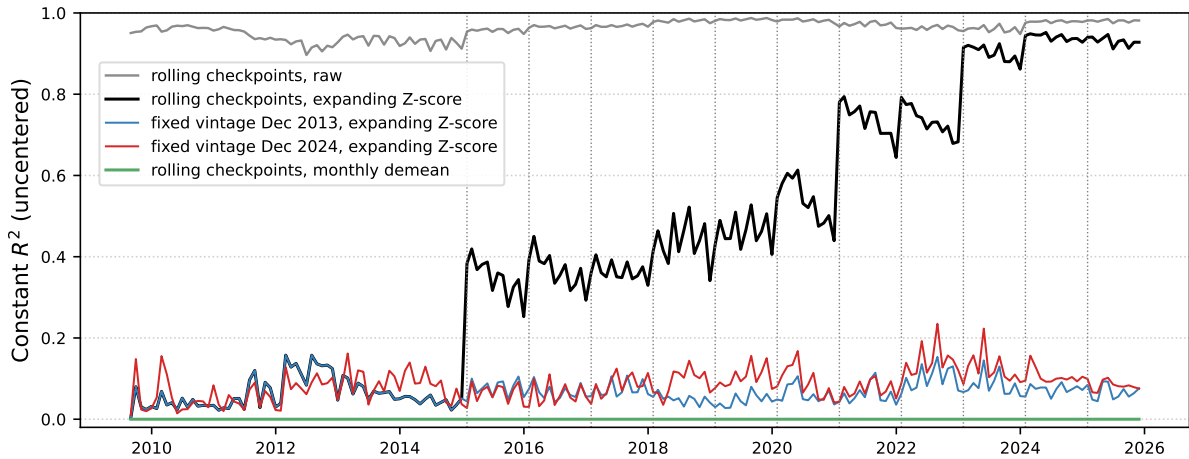


Figure 3: The Common Component Under Rolling Checkpoints

Note. This figure shows the monthly uncentered R^2 of the intercept-only fit, pooled across stocks and coordinates within each month: the share of the embeddings carried by the cross-sectional mean, on the broad universe. Dotted verticals mark the January checkpoint switches of the rolling scheme. The fixed-vintage series rebuild the same stock-month cells from embeddings extracted with a single checkpoint.

The cause is the checkpoint rotation. Each vintage places the centroid of its embeddings at its own location in the embedding space, so the centroid moves at every switch. An expanding standardization, estimated on the past, absorbs the new location only with a lag; the monthly cross-sectional demean subtracts the current month’s own mean and is immune by construction.

Figure 4 reports how much of the cross-sectional content the characteristics explain after the demean. The share rises with the richness of the residualization set, from essentially zero under *CAPM* to 6.6% under the full *JKP* set on the broad universe, close to the 7.7% that *DKPT* report on premium data. On the Nasdaq-100 the shares are uniformly smaller, at most 2.1%; the full *JKP* set cannot be estimated there, the monthly cross-section falling below its 130 regressors.

The characteristics thus span a predictable component of the news content; the residuals of these regressions, one per conditioning set, are the pure news embeddings. Whether removing that component improves the news portfolios is the question we ask.

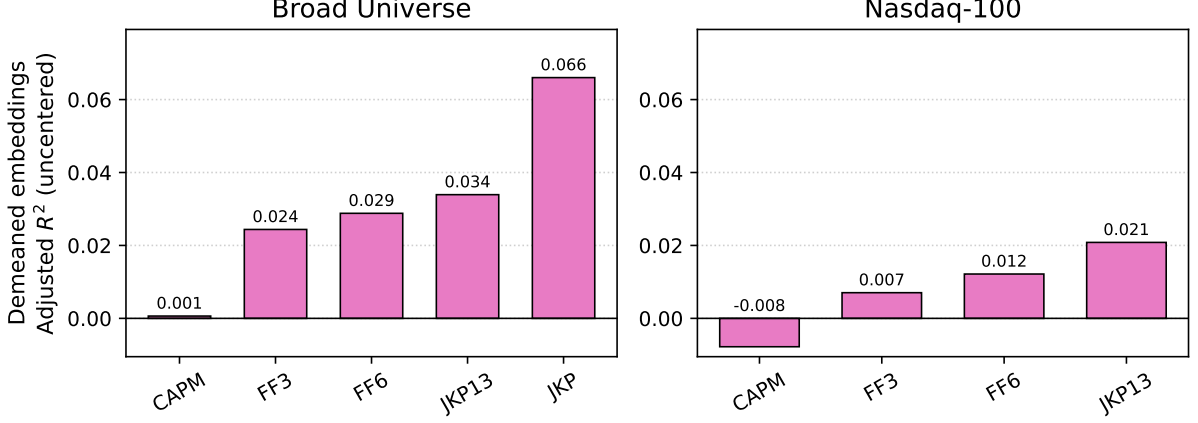


Figure 4: Predicting News Embeddings from Firm Characteristics

Note. This figure shows the pooled adjusted uncentered R^2 from monthly cross-sectional regressions of the demeaned news embeddings on the firm characteristics of each residualization specification, on both universes. After the monthly cross-sectional demean, every coordinate has mean zero by construction, so the uncentered denominator coincides with a within-month centered one. The *Constant* column is not shown, its fit being zero by identity on demeaned input; negative value reflect the degrees-of-freedom adjustment.

8 Portfolio Construction

Motivation. Instead of predicting returns and then sorting stocks on the forecasts, we estimate the portfolio weights directly with the penalized Maximum Sharpe Ratio Regression (MSRR) (Kelly and Xiu, 2023). The objective is risk-adjusted performance itself, so expected return and risk are handled jointly within a single estimation rather than in separate stages, while the ridge penalty mitigates overfitting in our high-dimensional setting.

Base Portfolios. The strategy is built from P base portfolios, one per embedding dimension. Writing $x_{t,s} \in \mathbb{R}^P$ for the month- t embedding of stock s used as the signal, each of its P dimensions induces a cross-sectional portfolio (stocks with higher values on that dimension receive larger weights), and we collect the P base-portfolio returns at $t + 1$ in

$$F_{t+1} = \frac{1}{|\mathcal{S}_t|} \sum_{s \in \mathcal{S}_t} x_{t,s} R_{t+1,s} \in \mathbb{R}^P,$$

where $\mathcal{S}_t$ is the set of stocks of the investable universe with a defined monthly embedding at t and a next-month return, and $R_{t+1,s}$ is that next-month excess return.

Because the base-portfolio construction is linear in the signal, the three embedding signals of Section 5 map into three families of portfolios, F^E , $F^{E|S}$, and F^ϵ , with $F^E = F^{E|S} + F^\epsilon$. We write F generically below.

MSRR. Penalized MSRR combines the base portfolios through a weight vector estimated to maximize a regularized Sharpe ratio. We use an expanding window: for each month $t \geq T$, with initial length $T = 12$, we estimate λ on months 1 to t and form the out-of-sample return at $t + 1$. For each value z of a fixed shrinkage grid, the MSRR weights solve

$$\hat{\lambda}_t(z) = \arg \min_{\lambda \in \mathbb{R}^P} \frac{1}{t} \sum_{u=1}^t (1 - \lambda^\top F_u)^2 + z_{\text{eff},t}(z) \|\lambda\|_2^2,$$

where the unit target encodes the maximum-Sharpe objective, and the effective penalty $z_{\text{eff},t}(z)$ rescales the shrinkage parameter z to the data:

$$z_{\text{eff},t}(z) = \frac{P}{t} z \operatorname{tr} \left(\frac{1}{t} \dot{\mathbf{F}}_t^\top \dot{\mathbf{F}}_t \right), \quad z \in \mathcal{Z} = \{10^{-6}, 10^{-5}, \dots, 10^5, 10^6\},$$

where P is the number of base portfolios and $\dot{\mathbf{F}}_t = (F_1, \dots, F_t)^\top \in \mathbb{R}^{t \times P}$ stacks the monthly base-portfolio return vectors, one row per month. The trace term is the total second moment of the base-portfolio returns over the window.¹⁰

Each grid value z thus defines a strategy, with out-of-sample return

$$r_{t+1}^{\text{MSRR}}(z) = \hat{\lambda}_t(z)^\top F_{t+1}.$$

We rescale each grid strategy ex post to a 10% target volatility by a constant factor $c(z)$,¹¹ and equally weight the grid:

$$\begin{aligned} r_{t+1}^{\text{ens}} &= \frac{1}{|\mathcal{Z}|} \sum_{z \in \mathcal{Z}} c(z) r_{t+1}^{\text{MSRR}}(z) = \frac{1}{|\mathcal{Z}|} \sum_{z \in \mathcal{Z}} c(z) \hat{\lambda}_t(z)^\top F_{t+1} \\ &= \underbrace{\left[\frac{1}{|\mathcal{Z}|} \sum_{z \in \mathcal{Z}} c(z) \hat{\lambda}_t(z) \right]^\top}_{\bar{\lambda}_t} F_{t+1} = \bar{\lambda}_t^\top F_{t+1}, \end{aligned}$$

¹⁰The shrinkage grid and the z_{eff} scaling differ between the printed equations of [KMSX](#) and the released code. The printed equations use the grid $\{10^{-6}, \dots, 10^{-1}, 1, 2, 5, 10, 10^2\}$ and include a $1/P$ factor in the trace term of z_{eff} (which cancels against the P/t prefactor, so the printed scaling does not depend on P); the code applies the decade grid $\{10^{-6}, \dots, 10^6\}$ with no $1/P$ factor (`utils/ridge.py` at commit `fd1aeb1`, April 2026). We raised this discrepancy with the authors, who confirmed to us by e-mail that the reported results follow the code at this commit, which we therefore adopt.

¹¹The volatility used for rescaling is estimated over the full prediction sample, as in the released code of [KMSX](#); this is the only non-point-in-time element of the procedure. Section [10.3](#) reports an expanding, point-in-time variant and shows that the results are robust to this choice.

since F_{t+1} is common to all grid values.

The ensemble is therefore of the same form, with effective weight vector $\bar{\lambda}_t$. Substituting the definition of F_{t+1} , its return can be written stock by stock, $r_{t+1}^{\text{ens}} = \frac{1}{|\mathcal{S}_t|} \sum_{s \in \mathcal{S}_t} (\bar{\lambda}_t^\top x_{t,s}) R_{t+1,s}$: stock s enters the portfolio with weight $\bar{\lambda}_t^\top x_{t,s} / |\mathcal{S}_t|$.

This procedure is applied identically, and separately, to each signal family, yielding the raw embeddings portfolio $F^{\star E}$, the predictable news portfolio $F^{\star E|S}$, and the pure news portfolio $F^{\star \epsilon}$, the latter two for each residualization specification.

Evaluation. Constrained below by the earliest point-in-time checkpoint, the out-of-sample window spans 143 monthly rebalances from January 2014 to November 2025; all performance statistics are computed over this window, Sharpe ratios and alphas are annualized. Unless stated otherwise, alphas are reported against each universe’s benchmark, QQQ for the Nasdaq-100 and the Fama–French market factor for the broad universe; t -statistics use HC3 heteroskedasticity-robust standard errors.

9 Results

9.1 The Nasdaq-100

Figure 5 reports the performance of the news portfolio strategies on the Nasdaq-100. Panel A reads from left to right along the residualization ladder; the raw embeddings portfolio is shown as the horizontal green dashed line. Only the pure news portfolios (pink bars) are dollar-neutral: the raw and predictable news portfolios (purple bars) inherit the non-zero cross-sectional mean of the embeddings and carry a net dollar exposure.

The raw embedding portfolio posts a Sharpe ratio of 0.64, and the predictable news portfolios reproduce it to the second decimal at every specification. The pure news portfolios, our main object of interest, are negative throughout. Under the *Constant* specification, the predictable news component is the cross-sectional mean of the embeddings, identical for every stock in a given month. All stocks therefore enter the portfolio with the same weight: the *Constant* predictable news portfolio is, by construction, an equal-weighted portfolio of the covered universe¹² held at a time-varying size.

¹²On the Nasdaq-100, this is close to the index itself: over the out-of-sample window the analyzed cross-section averages about 88 members per month, and covers nearly the whole index in the last three years (Section 6).



Figure 5: News Portfolio Performance on the Nasdaq-100

Note. This figure shows the performance of the MSRR news portfolios on the Nasdaq-100. Panel A reports the annualized Sharpe ratios across residualization specifications. Panel B reports the annualized alpha versus QQQ with HC3 standard errors and 95% confidence intervals. Panel C reports cumulative returns for the five pure news portfolios, the raw embeddings portfolio and the JKP13 predictable news portfolio. All portfolios are (ex post) standardized to 10% annual volatility for comparison purposes.

The richer predictable news specifications coincide with *Constant*: the explained uncentered R^2 share of Section 7 is identical to three decimals at every conditioning set, so the characteristic layers add essentially nothing to the common component. The raw portfolio is almost indistinguishable from them all: the common component alone attains an uncentered R^2 of 97.6%, so each stock's raw embedding is almost entirely its month's cross-sectional mean. The returns confirm the chain: all six portfolios are pairwise correlated above 0.999. Hence, every strategy above zero in Panel A is that same trade.

Panel B makes the same point in alpha terms, now against the QQQ tradable index itself. Every point estimate is negative and no confidence interval excludes zero. Panel C translates this record into cumulative returns. After twelve years, every pure news portfolio shows a cumulative loss, ending between -19% and -33% ; four of the five spend essentially the entire period below zero, and the *JKP13* portfolio floats above the pack before converging to it. The raw and predictable

news portfolios, one and the same position, end around +100%.

The three panels deliver the same message: the only above-zero performance is an equal-weighted market position that earns no alpha against the index it approximates, and the pure news portfolios deliver negative cumulative returns at every specification considered.

9.2 The Broad Universe

Broadening the cross-section from the Nasdaq-100 to the broad universe reverses the picture. Figure 6 repeats the analysis in the same format. In Panel A, the raw embedding portfolio earns a Sharpe ratio of 0.18 and the predictable news portfolios between 0.15 and 0.17. The pure news portfolios start at 0.79 under *Constant*, peak at 0.82 under *CAPM*, and then decline monotonically to 0.59 under the full *JKP* set. As on the Nasdaq-100, the raw and predictable news portfolios coincide: they are pairwise correlated above 0.998, and the common component alone attains a pooled uncentered R^2 of 96.3% (Section 7).

In Panel B, the raw embedding and predictable news portfolios earn -4.5% to -4.8% a year against the market factor, significant at the 5% level, largely a property of the equal-weighted position rather than of the embeddings.¹³ The pure news portfolios earn positive alphas, significant at least at the 5% level at every specification; the point estimate rises from $+8.6\%$ under *Constant* to $+8.8\%$ under *CAPM*, then falls to $+5.8\%$ under the full *JKP* set, where the lower bound of the confidence interval approaches zero. Panel C translates the record into cumulative returns. At the common 10% volatility scale, the pure news portfolios return between $+91\%$ (*JKP*) and $+149\%$ (*CAPM*), against $+17\%$ for the raw embedding portfolio and $+15\%$ for the *JKP* predictable news portfolio. The pure news paths share a distinctive shape: they drift slowly for eight years and accumulate most of their gains after 2022. Part of this is risk, not risk-adjusted performance: the realized volatility of the series roughly quadruples after 2022, about 4% annualized before against 16% after.

Figure 6 thus establishes that on the broad universe the signal is real: embeddings extracted from an accessible news provider with the PIT-4B-FT model predict cross-sectional returns out of sample after residualization, with alphas against the market factor significant at least at the 5% level at every specification. However significance weakens as the residualization set grows: the

¹³A mechanical equal-weighted portfolio of the same universe, built without any embedding input, earns an alpha of -3.5% a year ($t = -2.60$) against the market factor over the same window.

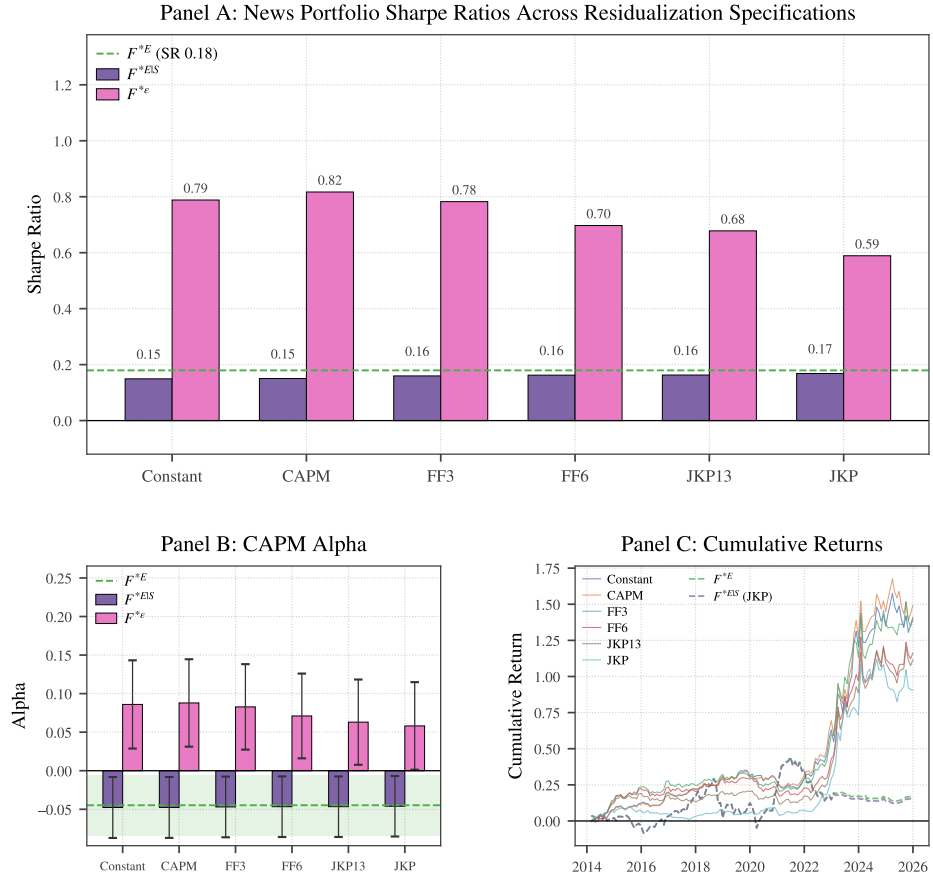


Figure 6: News Portfolio Performance on the Broad Universe

Note. This figure shows the performance of the MSRR news portfolios on the broad universe. Panel A reports the annualized Sharpe ratios across residualization specifications. Panel B reports the annualized alpha versus the market factor with HC3 standard errors and 95% confidence intervals. Panel C reports cumulative returns for the six pure news portfolios, the raw embeddings portfolio and the JKP predictable news portfolio. All portfolios are (ex post) standardized to 10% annual volatility for comparison purposes.

t -statistic falls from 2.9 under *Constant* to 2.0 under the full *JKP* set.

The early peak and decline of the pure news Sharpe ratios invert the premium-data ladder of [DKPT](#), which keeps rising, or at worst plateaus, toward much richer residualization sets. Yet the characteristics explain a similar share of the demeaned embeddings on both corpora.

9.3 Residualization Depth

Figures 5 and 6 leave a common question open. On the broad universe, the pure news Sharpe ratio falls as the set of controls grows richer; on the Nasdaq-100, it is negative at every specification and the ladder stops at *JKP13*, breadth being too thin to support the full 130 characteristic set. Neither pattern can be settled from five or six discrete specifications. The broad-universe decline may be a genuine property of the signal or an artifact of the particular characteristics used at

each specification. The Nasdaq-100 null may be robust to how much one residualizes, or may simply reflect an unlucky choice, with performance hidden at some richer specification.

Following [DKPT](#), Figure 7 traces, on both universes, the pure news Sharpe ratio as a function of the number of characteristics, k , from 0 to 100, used to residualize the embeddings. Each draw selects k characteristics at random from the full set of 130, residualizes the embeddings on them, and runs the MSRR estimation to obtain the pure news portfolio and its out-of-sample Sharpe ratio. We repeat this one hundred times for every k and, for each, plot the mean Sharpe ratio (the curve) and the 5th–95th percentile across draws (the band). Averaging over random draws washes out any performance specific to a particular choice of characteristics, so that the curve isolates the relationship between the amount of residualization and performance.

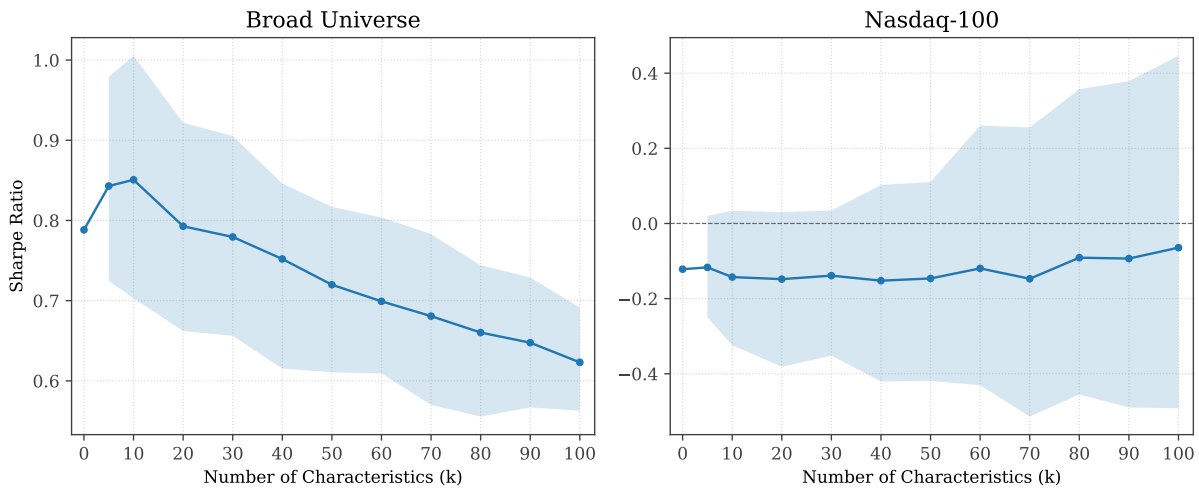


Figure 7: Pure News Sharpe Ratios and the Number of Characteristics

Note. This figure shows the annualized Sharpe ratio of the pure news portfolio against the number of characteristics k used to residualize the embeddings, on both universes. For each $k > 0$ the curve is the mean over 100 random subsets of k characteristics drawn from the 130 JKP characteristics; the shaded band is the 5th–95th percentile across draws. The point at $k = 0$ is the cross-sectional demean. The vertical scale differs between panels.

On the broad universe, the curve rises modestly from 0.79 at $k = 0$ to a peak of 0.85 near $k = 10$, then declines smoothly and without reversal to 0.62 at $k = 100$, and to 0.59 at the deterministic full set of 130, our *JKP* pure news specification. The decline spans some ninety values of k , so the gradient across our pure news specifications is not an artifact of the particular characteristics chosen at each step. The figure starts at the cross-sectional demean; the raw portfolio sits far below it, at a Sharpe ratio of 0.18. On our corpus, the gain thus comes entirely from that first step: conditioning on characteristics beyond it adds little at best, and costs performance at the deeper sets.

Under their Mistral-based extractor, [DKPT](#) report a Sharpe ratio that rises concavely in k without

reversal:¹⁴ purging characteristic content strengthens their pure news portfolio where it weakens ours. Nor is the rising shape a property of their flagship corpus alone: the third-party news aggregated by Reuters, which they exclude from their main sample on quality grounds, still rises across the ladder, reaching a Sharpe ratio of 2.3 at the full set against more than 3 for the main sample. Within their data, corpus quality shifts the level of the ladder, not its direction. A stricter comparison, against the point-in-time version of their results and under a closely aligned configuration, is deferred to Section 9.6.

On the Nasdaq-100, the curve is negative for every k up to 100: the null is robust to the amount of residualization, and no choice of k uncovers a signal on this universe.¹⁵

9.4 Factor Exposures Along the Residualization Ladder

Having traced the Sharpe ratio along the residualization path, we now turn to the loadings and the alphas. The alphas reported so far are measured against the market factor alone, so the first question is whether the performance identified at each rung survives richer controls. We therefore regress the pure news portfolio of each specification on the Fama–French six factors (FF5 + Mom) and on the market factor together with the thirteen JKP theme portfolios.

Table 2 reports the exposures to the Fama–French six factors. Under *Constant*, a fifth of the news portfolio’s variation is explained by the factors ($R^2 = 0.21$), with significant exposure to profitability (RMW 0.28, $t = 3.09$) and, more marginally, to momentum (0.22, $t = 1.89$) and against size (SMB -0.21 , $t = -1.77$). Despite this, the portfolio earns an alpha of 5.4% per year at 10% volatility ($t = 2.16$). As the predictable content is purged down the ladder, the exposures fade and the explained share falls with them, to 0.11 under *FF3* and 0.06 under *FF6*, where it settles: under *JKP*, no loading is distinguishable from zero, and the portfolio returns are largely decorrelated from the factors. The alpha, however, does not follow the exposures: it sits in a narrow band across all six specifications, 4.7 to 6.1% per year. It clears the 5% level down to *FF6*, but only the 10% level at the two deepest rungs, with $t = 1.91$ under *JKP13* and 1.74 under

¹⁴They report $SR(k)$ only for the Mistral extractor; for their point-in-time model, only the discrete ladder is available.

¹⁵At high k the residualization degenerates, the number of characteristics approaching the rank of the monthly characteristic matrix, itself bounded by the cross-section. The drift toward zero and the widening band are therefore numerical, not economic, and the full set ($k = 130$) is degenerate throughout, with a nominal $SR = +0.42$; it is excluded.

Table 2: Pure News Portfolio Exposures to the Fama–French Six Factors

This table reports alphas and betas of the pure news portfolio on the Fama–French six-factor model, across residualization specifications. All portfolios (both the dependent series and the factors) are (ex post) standardized to 10% annual volatility, and alphas are reported in annualized terms. t -statistics (HC3) are reported in parentheses; $n = 143$ monthly observations. ***, **, and * denote significance at the 1%, 5%, and 10% levels.

	<i>Constant</i>	<i>CAPM</i>	<i>FF3</i>	<i>FF6</i>	<i>JKP13</i>	<i>JKP</i>
Alpha	0.054** (2.16)	0.056** (2.27)	0.061** (2.35)	0.057** (2.16)	0.052* (1.91)	0.047* (1.74)
Market	0.047 (0.59)	0.055 (0.70)	−0.005 (−0.05)	−0.003 (−0.03)	0.056 (0.60)	0.006 (0.07)
SMB	−0.207* (−1.77)	−0.182 (−1.55)	−0.019 (−0.15)	−0.031 (−0.22)	−0.023 (−0.15)	−0.064 (−0.42)
HML	0.226 (1.53)	0.217 (1.48)	0.105 (0.72)	0.066 (0.43)	0.047 (0.29)	−0.068 (−0.43)
RMW	0.280*** (3.09)	0.293*** (3.18)	0.297*** (3.11)	0.190** (2.07)	0.161* (1.66)	0.149 (1.50)
CMA	−0.078 (−0.71)	−0.066 (−0.61)	−0.115 (−1.08)	−0.174 (−1.66)	−0.181 (−1.61)	−0.121 (−0.98)
Mom	0.215* (1.89)	0.220* (1.92)	0.167 (1.34)	0.095 (0.72)	0.063 (0.45)	−0.010 (−0.08)
R^2	0.209	0.205	0.109	0.056	0.048	0.055

JKP.¹⁶ The pure news portfolio thus does carry alpha relative to the six factors, on the order of 5% per year at 10% volatility, marginally so at the deepest rungs; whether it survives a richer benchmark is the question the JKP theme portfolios address.

Table 3 repeats the exercise against the market and the thirteen JKP anomaly theme portfolios. Under *Constant*, the themes explain a larger share of the portfolio’s variation ($R^2 = 0.30$), with a single exposure distinguishable from zero, on the size theme (−0.54, $t = -2.98$); down the ladder the explained share falls to 0.10, and under *JKP* no loading remains distinguishable. The alphas, however, no longer reach significance at any residualization depth: they sit between 3.9 and 4.9% per year, with t -statistics between 1.17 and 1.58.¹⁷ Hence against the theme portfolios, the pure news alpha of our accessible news provider is indistinguishable from zero at every residualization level.

¹⁶A paired difference regression of $r^{JKP} - r^{Constant}$ on the six factors confirms the stability: the difference in alphas is −0.6% per year ($t = -0.24$).

¹⁷A paired difference regression of $r^{JKP} - r^{Constant}$ on the market and the thirteen themes confirms the stability: the difference in alphas is +0.5% per year ($t = 0.18$).

Table 3: Pure News Portfolio Exposures to the JKP Theme Portfolios

This table reports alphas and betas of the pure news portfolio on the market factor and the 13 JKP theme portfolios, across residualization specifications. All portfolios (both the dependent series and the factors) are (ex post) standardized to 10% annual volatility, and alphas are reported in annualized terms. t -statistics (HC3) are reported in parentheses; $n = 143$ monthly observations. ***, **, and * denote significance at the 1%, 5%, and 10% levels.

	<i>Constant</i>	<i>CAPM</i>	<i>FF3</i>	<i>FF6</i>	<i>JKP13</i>	<i>JKP</i>
Alpha	0.039 (1.28)	0.039 (1.28)	0.049 (1.47)	0.039 (1.17)	0.039 (1.19)	0.044 (1.58)
Market	0.137 (1.15)	0.151 (1.25)	0.075 (0.60)	0.135 (1.05)	0.144 (1.08)	−0.001 (−0.01)
Accruals	−0.133 (−1.16)	−0.117 (−1.03)	−0.146 (−1.21)	−0.136 (−1.12)	−0.107 (−0.86)	0.008 (0.05)
Debt Issuance	0.066 (0.54)	0.105 (0.83)	0.077 (0.58)	0.024 (0.18)	0.064 (0.50)	0.062 (0.43)
Investment	0.137 (0.56)	0.104 (0.42)	−0.083 (−0.30)	−0.013 (−0.05)	−0.132 (−0.48)	−0.263 (−0.97)
Low Leverage	0.501 (0.87)	0.494 (0.85)	0.301 (0.44)	0.482 (0.70)	0.317 (0.52)	0.205 (0.41)
Low Risk	0.205 (0.65)	0.220 (0.69)	0.109 (0.30)	0.288 (0.79)	0.185 (0.56)	0.004 (0.02)
Momentum	0.149 (1.14)	0.162 (1.24)	0.245* (1.72)	0.228 (1.49)	0.231 (1.48)	0.207 (1.24)
Profit Growth	0.018 (0.12)	−0.001 (−0.01)	−0.079 (−0.53)	−0.086 (−0.52)	−0.121 (−0.69)	−0.224 (−1.31)
Profitability	−0.235 (−0.66)	−0.251 (−0.69)	−0.125 (−0.32)	−0.159 (−0.40)	−0.190 (−0.47)	−0.051 (−0.14)
Quality	−0.004 (−0.01)	0.013 (0.04)	0.048 (0.15)	0.013 (0.04)	0.054 (0.14)	0.058 (0.17)
Seasonality	0.005 (0.05)	−0.007 (−0.07)	−0.044 (−0.41)	−0.064 (−0.57)	−0.059 (−0.51)	0.053 (0.51)
Short-Term Reversal	0.069 (0.54)	0.065 (0.51)	0.038 (0.26)	0.107 (0.67)	0.135 (0.83)	0.186 (1.05)
Size	−0.542*** (−2.98)	−0.531*** (−2.84)	−0.239 (−1.16)	−0.138 (−0.69)	−0.122 (−0.61)	−0.135 (−0.62)
Value	0.589 (1.01)	0.622 (1.07)	0.516 (0.77)	0.465 (0.68)	0.445 (0.72)	0.349 (0.70)
R^2	0.297	0.297	0.166	0.113	0.098	0.095

Our Table 3 is the direct analogue of Table 4 in DKPT. On the exposure layer, their explained share falls from about 40% under *Constant* to 11% at the full purge, ours from 30% to 10%, though at their deep end significant exposures to low risk ($0.48, t = 2.00$) and short-term reversal ($0.15, t = 2.26$) emerge where ours retains none. On the alpha layer, however, the two corpora could not differ more, on two counts. At the starting point already: under *Constant*, before any characteristic purge, their news portfolio earns 14.6% per year net of the themes ($t = 7.6$), where ours earns nothing distinguishable from zero (3.9%, $t = 1.28$). And along the ladder: purging strengthens theirs gradually, up to 29.4% at the full characteristic set ($t = 12.7$), while ours stays flat, between 3.9 and 4.9%, indistinguishable from zero throughout. On premium news, the alpha against the themes is there before any characteristic is purged and grows as the purge deepens; on our accessible corpus, it is indistinguishable from zero before and after.

Taken together with the Sharpe profile of Figure 7, three facts summarize the ladder. Its one large gain is the demean; every layer beyond removes factor exposure and the compensation attached to it; and the alpha neither grows nor dies: against the six factors it clears the 5% level down to *FF6* and only the 10% level at the two deepest rungs, while against the themes it falls short throughout. A signal is nonetheless present at every depth: the pure news Sharpe ratios are positive and the alpha against the market factor runs from 8.6% ($t = 2.94$) to 5.8% ($t = 2.01$) per year (Panel B of Figure 6).

DKPT interpret pure news as the content of the text that is unexpected relative to the residualization set, and its performance as the market's delayed reaction to that content. On our corpus, that interpretation finds little to explain: deeper purges uncover no additional performance, so either the component orthogonal to stock characteristics carries less of the information that re-values assets than its premium counterpart, or the market absorbs most of it without delay. Under either reading, the news anomaly as they document it does not extend to our accessible corpus, even though a genuine signal remains against lighter benchmarks.

Two things differ between the two settings: the corpus and the implementation. On the corpus: theirs consists of articles carrying a body; a quarter of ours carries none, and restricting to body-carrying items would change coverage, which Section 9.5 shows drives performance and Section 9.9 quantifies. On the implementation: our pipeline follows KMSX, whose released code and conventions we can inspect, rather than DKPT, several of whose choices are undocumented or differ from ours. Section 9.6 makes the comparison under a closely aligned configuration, and

Section 9.7 shows that our conclusions do not depend on the estimator

9.5 Firm Size and Pure News Portfolio Performance

The pure news portfolio has so far failed on the Nasdaq-100. Before turning to alternative estimators, we ask what might explain the failure. On premium data, [DKPT](#) document a pronounced size asymmetry: the pure news portfolio performance is markedly weaker among large and mega caps (*Mega+Large*) than among small and micro caps (*Small+Micro*) or in the full universe. Whether that asymmetry holds in our setup, and what drives it, bears directly on the Nasdaq-100, whose members all belong to Mega+Large and are increasingly concentrated in the mega-cap tier.

Figure 8 shows it does, across every pure news specification: within Small+Micro the pure news portfolio earns a Sharpe ratio between 0.64 and 0.90, while within Mega+Large it is essentially absent, confined to a narrow band around zero (-0.09 to 0.16). The Small+Micro Sharpe ratio declines under heavier residualization, consistent with the broad universe results. Mega+Large falls short not only of Small+Micro but of the broad universe as a whole; on *JKP* its Sharpe ratio of 0.16 sits below the 0.59 of the full universe.

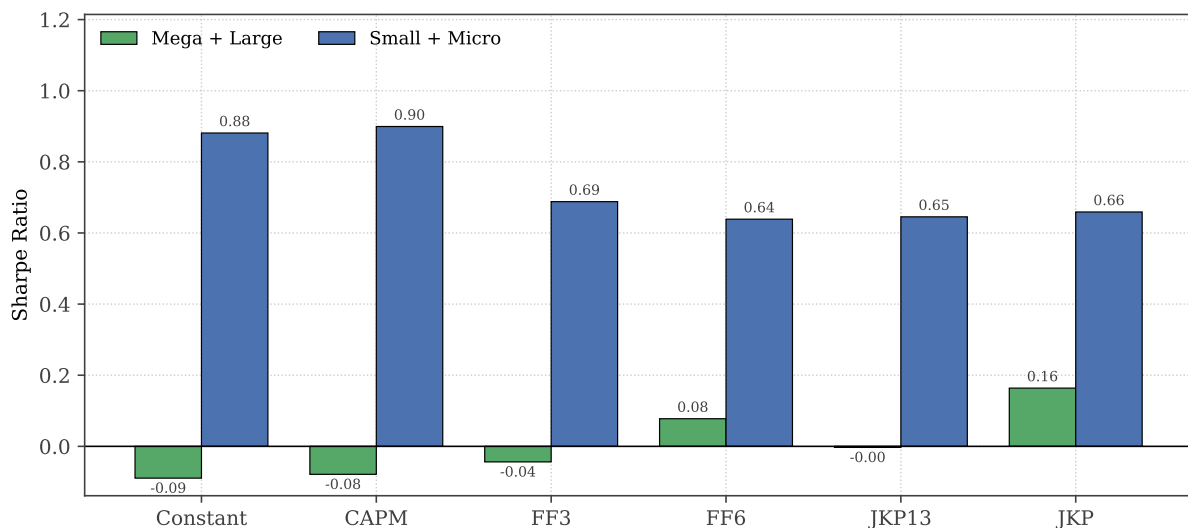


Figure 8: Pure News Sharpe Ratios by Size Group

Note. This figure shows the annualized Sharpe ratio of the pure news portfolio across residualization specifications, estimated within each of two size groups that partition the broad universe: Mega + Large and Small + Micro. For each group, the entire pipeline, cross-sectional residualization and MSRR estimation, is re-run within that group alone.

[DKPT](#) attribute this underperformance to breadth, through less diversification, rather than composition. The same explanation is available here: over the out-of-sample period, Mega+Large

averages only 937 stocks per month against 2,334 for the broad universe, and the Nasdaq-100 only 88. How much of the Mega+Large shortfall breadth accounts for is an empirical question.

Following [DKPT](#), we ask how the Sharpe ratio of the *JKP* pure news portfolio varies with the size of the cross section (N), and extend the exercise to the *Constant* and *JKP13* specifications. For each N on a grid from 300 to 2,000, we resample the cross section every month, keeping N names drawn uniformly from the broad universe, and re-estimate the entire pure news portfolio (both the cross-sectional residualization and the MSRR weights) on the drawn names, treating each month's draw as the investable universe. We repeat this 100 times per N and average the resulting Sharpe ratios. Under *Constant* and *JKP13* the grid extends down to 50 names, since those specifications remain estimable on cross sections too small to support the 130 characteristics of *JKP*. Because firms are drawn at random, the sampled cross sections match the composition of the broad universe in expectation each month, so the curve traces the Sharpe ratio as a function of N alone.

Figure 9 reports the result. In every specification the Sharpe ratio rises with the number of stocks drawn, from near zero at the smallest cross sections to its full-universe level. Breadth is a first-order determinant of the pure news portfolio's performance, as [DKPT](#) find on premium data. Under *JKP*, the Mega+Large portfolio lies close to the bootstrap mean at that breadth. At the same number of stocks, Mega+Large therefore performs close to what the composition of the broad universe delivers: its shortfall appears to be a breadth effect rather than evidence that news predicts returns less well within the group, replicating the conclusion [DKPT](#) reach for their Mega+Large subsample.

Extending the bootstrap to the shallower specifications gives a less clean reading. At those depths Mega+Large sits below the breadth-matched curve, suggesting that breadth might not be the only factor behind its underperformance relative to the broad universe; the gap narrows from both sides as residualization deepens, the curve falling while Mega+Large rises. The reading is asymmetric: a point on the curve is consistent with breadth alone, while a point away from it admits several readings. Separating them would require a formal test we do not pursue, since our conclusions for the Nasdaq-100 do not depend on it.¹⁸

¹⁸At its own breadth (1,398 stocks), Small+Micro lies above the bootstrap curve in every specification; the same caveat applies.

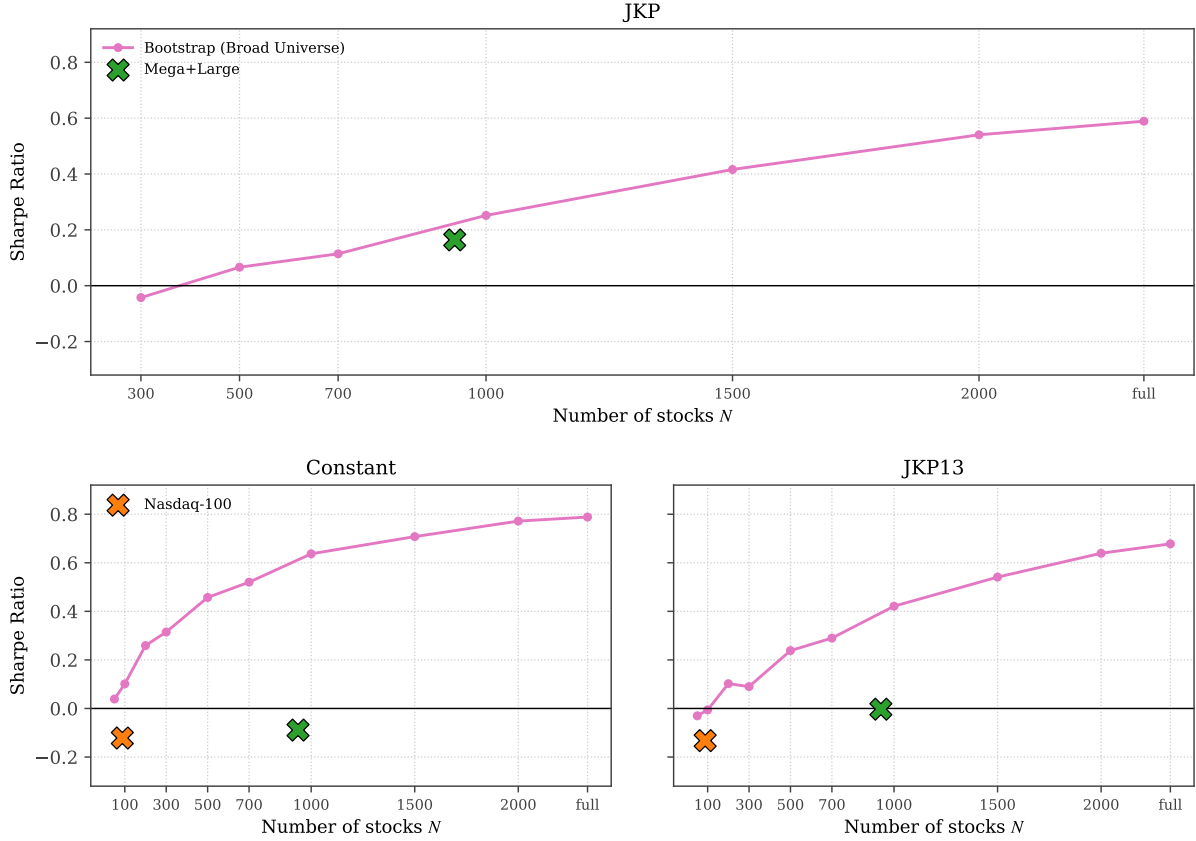


Figure 9: Pure News Sharpe Ratios and Cross-Sectional Breadth

Note. This figure shows the annualized Sharpe ratio of bootstrapped pure news portfolios against the number of stocks N drawn from the broad universe each month, with the cross-sectional residualization and the MSRR weights re-estimated on every draw and averaged over 100 bootstrap samples. We repeat it for *JKP*, *Constant*, and *JKP13*. The last point of each curve is the full broad universe (2,334 stocks on average out of sample). The green cross marks the pure news portfolio estimated within Mega + Large and the orange cross the portfolio estimated on the Nasdaq-100, each placed at its own out-of-sample mean breadth (937 and 88 stocks).

The Nasdaq-100 is exposed on both counts: it belongs to the group that appears penalized relative to the broad universe (we can establish neither why nor whether the difference is significant), and its breadth falls in the range where the bootstrap curves return approximately zero. Either way, a cross section of that size leaves little room for the pure news portfolio. This motivates the alternative constructions examined in Sections 9.7 and 9.8.

9.6 The Premium Gap

In this section we quantify *The Premium Gap*: the performance cost of substituting an accessible news provider for the premium data on which the literature's headline results are established. The benchmarks are built on premium newswires: Dow Jones used by [KMSX](#) and Reuters used by [DKPT](#).

A naive comparison of headline Sharpe ratios cannot answer this question: beyond the news data itself, our results and theirs differ in the sample window and, in the case of [DKPT](#), in several methodological choices. Our design therefore minimizes these confounders by re-running our entire pipeline under conditions closely aligned with each benchmark, while leaving the constraints of our own provider in place. For every comparison, the sample is truncated at the end of the benchmark’s data: May 2020 for [KMSX](#) and December 2022 for [DKPT](#). Against [KMSX](#), no further alignment is needed. Against [DKPT](#), we additionally adopt their article-mean monthly aggregation, expanding Z-score of the embeddings, leave-one-out selection of the MSRR shrinkage, and their definition of the *JKP* set with 132 characteristics instead of 130.

Table 4: The Premium Gap: The Price of Accessible News

This table shows the annualized pure news Sharpe ratios on the broad universe, computed to each benchmark’s data end, with each block aligned with its benchmark’s configuration. The Premium Gap column reports the benchmark Sharpe ratio minus ours. References: PIT-4B-FT from Figure 3 of [KMSX](#) (reported for *JKP* only); PIT-1.5B from Figure 20 of [DKPT](#); values are read from the published figures. The *JKP* set contains 130 characteristics in the [KMSX](#) block and 132 in the [DKPT](#) block.

	KMSX			DKPT		
	Dow Jones, to 2020:05			Reuters, to 2022:12		
	Ref	Ours	Premium Gap	Ref	Ours	Premium Gap
<i>Constant</i>	—	0.76	—	1.35	0.79	0.56
<i>CAPM</i>	—	0.74	—	1.50	0.79	0.71
<i>FF3</i>	—	0.63	—	1.80	0.78	1.02
<i>FF6</i>	—	0.56	—	1.95	0.60	1.35
<i>JKP</i>	1.53	0.19	1.34	1.90	0.49	1.41

Table 4 reports the resulting Sharpe ratios alongside the benchmark values. Two results emerge. First, the gap is positive and pervasive: everywhere a benchmark reference exists, our aligned portfolio sits below it. Second, its magnitude deepens with residualization depth. At the shallowest rung the gap is smallest: our constant-specification portfolio reaches 0.79 against the [DKPT](#) benchmark of 1.35, retaining close to 60% of the premium level. Moving down the ladder, the gap widens from 0.56 to 1.41, and the retained share falls to about a quarter. The gap opens from both sides: the [DKPT](#) benchmark strengthens as the conditioning set grows while our portfolios weaken, the pattern documented in Section 9.3. Their Mistral-extracted pure news portfolio keeps rising through the full *JKP* set; with their point-in-time model the ladder saturates at *FF6*. Failing to rise all the way is therefore not specific to our corpus, though ours does not merely level off, it declines.

At the *JKP* rung, the same picture holds against the second benchmark: 0.19 against 1.53 in the

[KMSX](#) panel, barely more than a tenth of the premium level. This deepest rung, the only one for which both benchmarks report a reference, is where the gap is corroborated by two distinct premium providers, Dow Jones and Reuters. The Premium Gap is thus not a uniform level difference: it concentrates at the deepest conditioning levels, precisely the specifications that produce the literature’s headline results, and those at which the news anomaly was shown not to extend to our corpus.

The datasets do not begin at the same date, even though the evaluation windows coincide. The benchmarks enter the out-of-sample window with MSRR weights estimated on thirty additional years of history for [KMSX](#) and thirteen for [DKPT](#), a potential advantage that persists at every out-of-sample date. This asymmetry, however, is not a flaw of the comparison; it is part of what is being compared. As documented in Section 3, the lack of historical depth is a common feature of accessible news providers. Nor is it the only one. An accessible provider differs from a premium newswire along several dimensions at once, editorial resources and original reporting, but also coverage, article format, and tagging quality; these are joint features of the tier, not separable treatments. We make no attempt to decompose the gap into these channels: what the table prices is the substitution of one package for the other, which is also the decision an investor or researcher actually faces.

However, breadth, the first-order determinant documented in Section 9.5, does not appear to confound the comparison, at least against [DKPT](#): their cross section averages 4,198 stocks per month, of which 52.5% carry news, so roughly 2,200 news-carrying names against our 2,221 over the matched window. Their figure is a full-sample average that we cannot restrict to 2014–2022, so the comparison is approximate; the two are of the same order, which is all the argument requires.¹⁹

Two caveats bound the interpretation, both outside the package itself. The [DKPT](#) reference is produced by the PIT-1.5B model, a smaller model than our PIT-4B-FT, so the right panel probably understates rather than overstates the gap. And part of their configuration is undocumented: the estimation conventions in the aligned column remain ours, from the shrinkage grid and its scaling to the normalization of the base portfolios. Whether their point-in-time embeddings are extracted by mean pooling, as their Mistral ones, or from the last token as in [KMSX](#), is not reported.

¹⁹[KMSX](#) do not report the size of their news universe.

9.7 Pooled Sorts under Squared-Error Selection

Section 9.5 established that the MSRR estimator is tightly dependent on the breadth of the investable universe: its performance scales with the number of stocks the cross-sectional optimizer can exploit, and on the narrow Nasdaq-100, pure news portfolio performance collapses. Following DKPT, who build on CKX, we replace the direct maximization of the Sharpe ratio by a two-step procedure: we first predict stock-level returns from the news embeddings, then form portfolios by sorting on the predictions. We investigate whether the change of estimator rehabilitates the news signal on the Nasdaq-100 and whether our broad-universe results are robust to it.

We replace the Sharpe-maximizing MSRR weights $\hat{\lambda}_t$ with a pooled ridge regression of one-month-ahead returns on the embeddings. Over the same expanding window, we pool every stock-month available before t and solve

$$\hat{b}_t(z) = \arg \min_{b \in \mathbb{R}^P} \frac{1}{N_t} \sum_{u < t} \sum_{s \in \mathcal{S}_u} (R_{u+1,s} - b^\top x_{u,s})^2 + z_{\text{eff},t}(z) \|b\|_2^2,$$

where $x_{u,s} \in \mathbb{R}^P$ is the month- u embedding of stock s , $N_t = \sum_{u < t} |\mathcal{S}_u|$ the number of pooled stock-month observations, and $z_{\text{eff},t}(z)$ rescales the grid shrinkage $z \in \mathcal{Z}$, with $\hat{\mathbf{F}}_t$ replaced by the pooled design matrix $\hat{\mathbf{X}}_t \in \mathbb{R}^{N_t \times P}$ stacking the $x_{u,s}$, and t by N_t . In contrast to the MSRR ensemble, we do not average across the grid: each month we retain the single shrinkage z that minimizes the leave-one-out prediction error on the pooled panel. The resulting coefficient vector $\hat{b}_t$ is common to all stocks and condenses the embedding of stock s into a scalar expected-return signal,

$$\hat{\mu}_{t,s} = \hat{b}_t^\top x_{t,s}.$$

We then rank the stocks in $\mathcal{S}_t$ on $\hat{\mu}_{t,s}$ and hold an equal-weighted long–short portfolio, long the top 20% and short the bottom 20%.

Panel A of Table 5 reports the sorts on the broad universe. Down to *JKP13*, the news signal survives the change of estimator: the high-minus-low spread earns a significant CAPM alpha throughout, with t between 2.69 and 3.71, and Sharpe ratios between 0.79 and 0.96. The spread narrows with residualization depth, but its volatility falls in step, so risk-adjusted performance does not deteriorate. The Low quintile earns the least, and average returns rise broadly across the bins: the predictions order the cross-section, not just the two extreme baskets.

Table 5: Pure News Portfolio Sorts Under MSE Selection

This table shows equal-weighted quintile portfolios formed each month by sorting stocks on the predicted return of the pooled ridge, with the ridge penalty selected by the MSE rule defined in the text, for each residualization specification. Panel A reports the broad universe, Panel B the Nasdaq-100. The first five columns report the annualized average return of each score quintile, in excess of the risk-free rate (%). The remaining columns report the high-minus-low spread: annualized mean (%), volatility (%), CAPM alpha (%) against the market factor in Panel A and against the Nasdaq-100 benchmark (QQQ, in excess of the risk-free rate) in Panel B, its HC3 t -statistic, and the annualized Sharpe ratio. $n = 143$ monthly observations.

Panel A: Broad Universe										
	Excess return by score quintile					High-minus-low spread				
	Low	2	3	4	High	Mean	Vol	α	$t(\alpha)$	SR
<i>Constant</i>	1.5	8.0	8.3	10.5	10.7	9.1	11.2	10.6	3.33	0.81
<i>CAPM</i>	1.2	8.4	8.2	9.8	11.6	10.3	10.8	11.4	3.71	0.96
<i>FF3</i>	3.5	7.4	8.7	9.5	10.0	6.5	8.1	7.5	3.34	0.80
<i>FF6</i>	3.4	8.0	8.2	10.3	9.3	5.9	7.3	6.3	3.12	0.80
<i>JKP13</i>	3.6	7.6	9.3	9.9	8.8	5.2	6.6	5.1	2.69	0.79
<i>JKP</i>	6.8	8.6	8.7	8.4	6.7	-0.1	4.1	-0.3	-0.23	-0.02

Panel B: Nasdaq-100										
	Excess return by score quintile					High-minus-low spread				
	Low	2	3	4	High	Mean	Vol	α	$t(\alpha)$	SR
<i>Constant</i>	11.4	13.2	14.9	14.3	11.7	0.2	11.0	-2.5	-0.83	0.02
<i>CAPM</i>	13.4	13.5	15.4	11.0	12.0	-1.3	10.9	-3.4	-1.14	-0.12
<i>FF3</i>	13.5	15.2	12.2	12.7	11.9	-1.6	10.6	-2.4	-0.78	-0.15
<i>FF6</i>	16.2	11.3	13.3	15.0	9.6	-6.6	9.3	-7.7	-2.88	-0.71
<i>JKP13</i>	13.7	12.8	12.8	13.6	12.5	-1.3	10.0	-1.4	-0.52	-0.13

At *JKP*, however, the sort collapses: the spread is indistinguishable from zero and the bins no longer order the cross-section. On the Nasdaq-100 (Panel B), the change of estimator rehabilitates nothing: no rung delivers a positive alpha, and no ordering emerges across the bins. The null documented under MSRR is thus not an artifact of that estimator. The selected penalties point to the cause of both failures. On the broad universe, the squared-error criterion selects a stable shrinkage in every out-of-sample month at every level down to *JKP13*. At *JKP*, it never selects that value, its choice scattering across four orders of magnitude. On the Nasdaq-100, the selection is unstable at every residualization level, and the penalties it picks are far heavier than on the broad universe. In both cases, the failure sits at the prediction stage, upstream of any portfolio construction: the regression finds no stable signal to estimate, and the sorts merely inherit that instability.

9.8 Alternative Shrinkage Selection Rules

Squared prediction error scores the levels of the predictions, while the composition of the sorted portfolio depends only on their ranks. We therefore consider three alternative ridge shrinkage selection rules, each scoring a candidate penalty on the quality of the ranking its held-out predictions produce within the training window, in the same expanding leave-one-out design. *Rank-IC* retains the penalty with the highest average cross-sectional Spearman correlation between held-out predictions and realized returns; the *Tile spread* retains the penalty with the largest average monthly return spread between the top and bottom quintiles of the held-out predictions; and the *Sharpe* criterion retains the penalty whose held-out long–short returns earn the highest Sharpe ratio. These rules align the selection of the penalty, in our pooled design, with the portfolio objective. Additionally, we report a quintile sort on the MSRR per-stock weights (Table C.1), so that every signal is compared through the same portfolio construction.

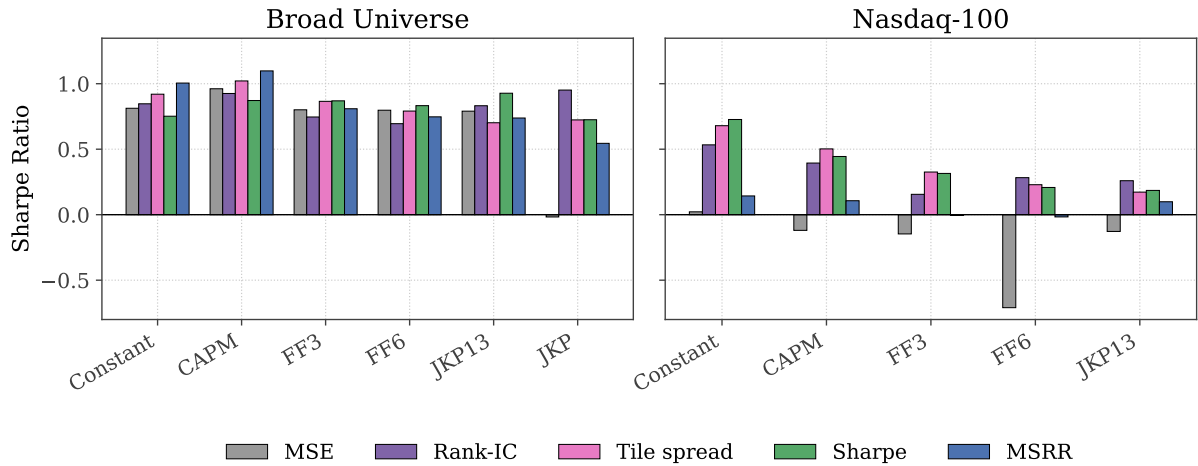


Figure 10: Pure News Sharpe Ratios Across Selection Rules

Note. This figure shows the annualized pure news Sharpe ratios of high-minus-low quintile spreads on both universes across residualization specifications. All bars share the same portfolio construction, an equal-weighted quintile spread on the same target. The first four bars are the pooled ridge under each selection rule: leave-one-out mean squared prediction error (MSE) and the three ranking-based rules defined in the text. The MSRR bar sorts stocks on the weights of the MSRR strategy.

Figure 10 summarizes the sorts under each selection rule. On the broad universe, the choice of rule is largely immaterial down to *JKP13*: all rules deliver similar Sharpe ratios. At *JKP*, the ranking-based rules revive the signal the squared-error criterion missed, with Sharpe ratios between 0.72 and 0.95, against 0.55 for the sort on the MSRR weights: the deepest rung does carry a signal, and the ranking rules capture it best. On the Nasdaq-100, where both the MSE rule and the sort on the MSRR weights deliver almost nothing, the ranking-based rules lift the Sharpe ratio to between 0.5 and 0.75 at *Constant*, fading as residualization deepens.

Table 6 details the sorts under the Sharpe rule, the most favorable configuration for the Nasdaq-100. On the broad universe (Panel A), the signal holds at every rung of the ladder: even at *JKP*, the weakest, the spread earns an alpha of 2.8% per year ($t = 2.08$), and the two other ranking rules deliver similar or stronger spreads at that rung (Tables C.2 and C.3).

Table 6: Pure News Portfolio Sorts Under Sharpe Selection

This table shows equal-weighted quintile portfolios formed each month by sorting stocks on the predicted return of the pooled ridge, with the ridge penalty selected by the Sharpe rule defined in the text, for each residualization specification. Panel A reports the broad universe, Panel B the Nasdaq-100. The first five columns report the annualized average return of each score quintile, in excess of the risk-free rate (%). The remaining columns report the high-minus-low spread: annualized mean (%), volatility (%), CAPM alpha (%) against the market factor in Panel A and against the Nasdaq-100 benchmark (QQQ, in excess of the risk-free rate) in Panel B, its HC3 t -statistic, and the annualized Sharpe ratio. $n = 143$ monthly observations.

Panel A: Broad Universe										
	Excess return by score quintile					High-minus-low spread				
	Low	2	3	4	High	Mean	Vol	α	$t(\alpha)$	SR
<i>Constant</i>	1.9	7.2	9.9	10.2	9.9	8.0	10.6	9.6	3.09	0.75
<i>CAPM</i>	1.6	7.4	8.8	10.6	10.6	9.0	10.3	10.4	3.38	0.87
<i>FF3</i>	3.4	7.6	8.0	10.0	10.2	6.8	7.8	7.3	3.27	0.87
<i>FF6</i>	3.8	8.0	8.6	9.2	9.6	5.8	7.0	6.0	3.16	0.83
<i>JKP13</i>	3.9	7.6	8.6	9.3	9.7	5.7	6.2	5.7	3.17	0.93
<i>JKP</i>	5.5	7.8	8.7	8.5	8.7	3.1	4.3	2.8	2.08	0.72

Panel B: Nasdaq-100										
	Excess return by score quintile					High-minus-low spread				
	Low	2	3	4	High	Mean	Vol	α	$t(\alpha)$	SR
<i>Constant</i>	9.4	11.2	12.6	15.0	17.2	7.7	10.6	5.7	1.87	0.73
<i>CAPM</i>	9.6	13.9	11.0	16.6	14.4	4.8	10.9	3.5	1.07	0.44
<i>FF3</i>	11.8	9.4	15.4	13.6	15.4	3.6	11.3	2.3	0.67	0.32
<i>FF6</i>	11.3	10.6	14.1	16.2	13.3	2.1	10.0	1.0	0.33	0.21
<i>JKP13</i>	12.0	15.8	12.9	10.7	13.9	1.9	10.0	1.8	0.61	0.19

On the Nasdaq-100 (Panel B), the recovery concentrates at *Constant*: the bins rise monotonically from 9.4% to 17.2%, the only ordered sort on the Nasdaq-100 in this section, and the spread earns an alpha of 5.7% per year, marginally significant ($t = 1.87$). The pattern fades from *CAPM* onward. Even under the most favorable rule, no alpha is significant at the 5% level, and only the *Constant* spread reaches the 10% level: the ranking-based rules recover a weak signal at the lightest residualization level, not a rehabilitation of the Nasdaq-100.

Taken together, the pooled exercise answers its two opening questions. The broad-universe signal is not an artifact of the MSRR machinery: the pooled ridge recovers it at every residualization level, the squared-error rule at *JKP* excepted, an exception that lies in the selection criterion rather than in the signal. The Nasdaq-100, by contrast, is improved but not rehabilitated. And under

either estimator and any selection rule, the news anomaly itself does not reappear: performance stays stable or declines at the deepest residualization levels rather than building along them.

9.9 News Input and Pure News Performance

The baseline embeds each news item’s headline and body, truncated at 2,048 tokens. In the broad universe, 26.8% of the items carry no body (21.1% on the Nasdaq-100), so the monthly average mixes two kinds of representation, a headline for some items and a headline followed by a narrative for others. We call an input free of that mixture *homogeneous*. This section asks which part of a news item carries the signal, and what role the mixture of representations and the breadth of the feed play in the answer. Two manipulations cross to that end.

The first is the news item set, either every item or only those carrying a body. The second is the embedded text, either the headline alone or the headline followed by the body. Crossing them yields four inputs. *Baseline* is one of them, every item with its body where it has one; the other three are *Headline-only*, every item reduced to its headline; *Full articles*, only the items carrying a body; and *Full articles-Headline*, those same items reduced to their headline.²⁰ We vary the input at fixed estimator: the canonical MSRR specification on the broad universe, and the pooled sort under Sharpe selection on the Nasdaq-100 (Section 9.8). Comparisons are therefore intra-universe, each panel holding its estimator fixed.

Figure 11 shows the results. Take the narrow universe first. On the Nasdaq-100 no ordering emerges: the four inputs cross along the ladder, each of them leading at some level, so this universe does not separate them. On the broad universe the ordering is the same at every residualization level: *Headline-only* dominates, with Sharpe ratios between 0.91 and 1.16; *Full articles-Headline* follows, between 0.61 and 1.01; then *Baseline*, between 0.59 and 0.82; and *Full articles* comes last, between 0.14 and 0.50. The two extremes, *Full articles* and *Headline-only*, are far apart, the Sharpe ratio at least doubling between them at every residualization level. But they differ in two things at once, the text embedded and the news items kept, so the gap could come from the text itself or from the change in breadth (Section 9.5). The four inputs allow the two channels to be inspected through paired comparisons.

²⁰We identify headline-only items by a blank body. The feed carries no field that would set alert-style items apart, and any length rule would be arbitrary (Section 4.2). Items whose body is not empty but nearly so, such as a bare source attribution (“-Reuters”), escape the blank-body test: the count is therefore a lower bound.

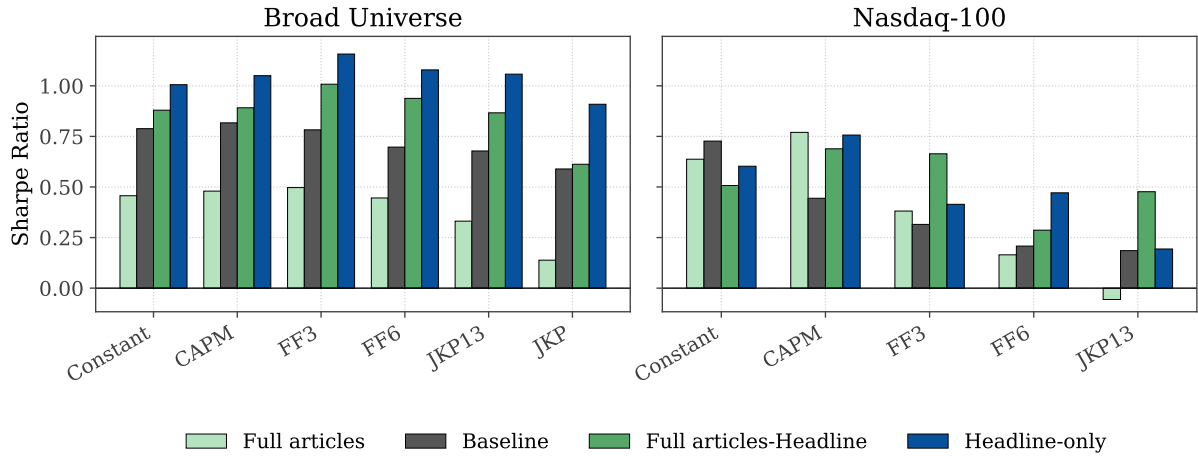


Figure 11: Pure News Sharpe Ratios Across News Inputs

Note. This figure shows the annualized pure news Sharpe ratios on both universes across residualization specifications, for the *Full articles*, *Baseline*, *Full articles-Headline* and *Headline-only* news inputs described in the text. The broad-universe strategy is the MSRR ensemble with continuous weights; the Nasdaq-100 strategy is the pooled ridge under the Sharpe selection rule, evaluated as the equal-weighted high-minus-low quintile spread.

The first two comparisons ask where the signal lives within an item. They hold the news items fixed and vary only the text: *Baseline* against *Headline-only* over every item, and *Full articles* against *Full articles-Headline* over the body-carrying items alone. In both, the headline version comes out ahead at every residualization level. Coverage being identical on the two sides, the gain has a single source: embedding the body degrades the monthly signal. The performance improvement from *Full articles* to *Full articles-Headline*, where both inputs are homogeneous, suggests that the gain from *Baseline* to *Headline-only* is not just an effect of removing the mixture of news representations. What remains is dilution inside the representation of the text itself. The headline is a condensed, impactful version of the information; the body, whether through context, background or templated narrative, dilutes the quality of the signal. This does not say the body carries no information, only that, embedded this way, it obscures more than it adds relative to the headline alone.

Two further comparisons ask whether the headline-only items belong in the feed, or whether it should be restricted to full articles only as [DKPT](#) do. *Full articles* against *Baseline* answers it directly: the two differ only in that restriction, and it costs performance at every residualization level. *Full articles-Headline* against *Headline-only* repeats the restriction under the headline text rule, with the same outcome: restricting costs performance whether the remaining items are embedded with their bodies or as headlines alone, through the signal the dropped items carry, the coverage they provide, or both.

A final comparison completes the reading. *Full articles-Headline* covers 18.4% fewer names per month than the *Baseline* (1,905 against 2,334 on average over the out-of-sample window) yet beats it at every level: when the two channels pull in opposite directions, removing the body outweighs the loss of coverage.

Table 7: Pure News Portfolio Alphas Under the Baseline and Headline-Only Inputs

This table reports annualized alphas from the pure news MSRR portfolios under the *Baseline* and *Headline-only* news inputs, across residualization specifications. Column headings are the pure news residualization specifications; each row is the alpha against one pricing model: the CAPM, the Fama–French six factors, and the market with the 13 JKP theme portfolios. Pure news portfolio returns are (ex post) standardized to 10% annual volatility. *t*-statistics (HC3) are reported in parentheses; $n = 143$ monthly observations. ***, **, and * denote significance at the 1%, 5%, and 10% levels.

	<i>Constant</i>	<i>CAPM</i>	<i>FF3</i>	<i>FF6</i>	<i>JKP13</i>	<i>JKP</i>
<i>Panel A: Baseline</i>						
CAPM	0.086*** (2.94)	0.088*** (3.04)	0.083*** (2.93)	0.071** (2.53)	0.063** (2.23)	0.058** (2.01)
FF6	0.054** (2.16)	0.056** (2.27)	0.061** (2.35)	0.057** (2.16)	0.052* (1.91)	0.047* (1.74)
Themes	0.039 (1.28)	0.039 (1.28)	0.049 (1.47)	0.039 (1.17)	0.039 (1.19)	0.044 (1.58)
<i>Panel B: Headline-only</i>						
CAPM	0.108*** (3.92)	0.110*** (4.03)	0.120*** (4.37)	0.109*** (4.00)	0.100*** (3.66)	0.082*** (2.86)
FF6	0.081*** (2.90)	0.085*** (3.04)	0.105*** (3.68)	0.100*** (3.39)	0.096*** (3.21)	0.080** (2.59)
Themes	0.073** (2.45)	0.074** (2.47)	0.085*** (2.76)	0.074** (2.42)	0.074** (2.44)	0.071** (2.16)

Table 7 compares the *Baseline* and *Headline-only* inputs in alpha terms against three factor models. Panel A restates the *Baseline* described in Section 9.4 for reference. Panel B reads differently. Every alpha is significant at least at the 5% level, for all three models and at all six residualization specifications; against the market, all clear the 1% threshold. The point estimates lie above their Panel A counterparts in all eighteen cells, close to twice as large at some levels.

Strong as it is, the headline-only input does not rehabilitate the news anomaly of [DKPT](#). There, the alpha rises gradually with the residualization set, from 14.6% at *Constant* to 29.4% at *JKP*, under the same theme-portfolio controls and the same 10% standardization; ours stays flat. Our best-performing input has, on the other hand, a practical corollary. Extraction cost scales with the volume of text embedded, and the title averages about 20 tokens per item against about 175

for the full input: embedding headlines alone is an order of magnitude cheaper. The saving is directly relevant to the audience an accessible provider serves, where embedding more than one million full items may be a binding cost.

9.10 Transaction Costs

The performance documented so far is gross of costs. This section quantifies the impact of transaction costs and measures net-of-cost performance for six strategies of the preceding sections: the *Constant* and *JKP* pure news MSRR portfolios on the broad universe, each under the *Baseline* and *Headline-only* inputs, and the *Constant* pooled Sharpe sort on each of the two universes.

On our corpus, the component of the embeddings orthogonal to the known risk factors does not deliver superior performance: individual intermediate levels can edge above *Constant*, but residualization beyond the cross-sectional mean has not paid off systematically, performance drifting down the ladder rather than building along it. We therefore take the lightest purge, the *Constant* pure news portfolio, as the implementation candidate, rather than cherry-picking ex post the intermediate level that happens to perform best. The *JKP* pure news portfolio is carried through as the configuration behind the literature’s headline claim, even though that claim does not replicate here. Among the selection criteria of Section 9.8 on the Nasdaq-100, we take the one with the highest alpha t -statistic; the choice is deliberately favorable, and even so the configuration falls short of significance at the 5% level before costs. Its broad-universe counterpart is included for symmetry.

We follow the conventions of [DKPT](#). One-sided turnover, the fraction of the book replaced at each rebalance, is one-half of the total absolute change in positions between the portfolio inherited from $t - 1$ after return drift and the portfolio rebalanced at t , relative to the previous month’s gross exposure,

$$\text{Turnover}_t = \frac{1}{2G_{t-1}} \sum_s |w_{s,t} - (1 + R_{t,s}) w_{s,t-1}|, \quad G_{t-1} = \sum_s |w_{s,t-1}|,$$

where $w_{s,t}$ is the position of stock s at formation month t , G_{t-1} the total gross exposure at the end of month $t - 1$, and $R_{t,s}$ the return earned over the month those previous positions were held. Costs are linear at 10 basis points per dollar traded, following [Frazzini et al. \(2018\)](#).²¹

²¹The calibration captures execution costs only: lending fees on the short leg are excluded.

Table 8 reports the results. Monthly one-sided turnover is between 0.74 and 0.82 for every strategy: roughly four fifths of the book’s value is replaced each month, whatever the estimator, the universe, or the input; the portfolio carries little over from one month to the next. The *JKP* pure news portfolio does not survive costs under either input: its net Sharpe ratio is negative. A stronger input does not rescue it, the *Headline-only* version turning negative just the same despite higher gross performance. Net Sharpe ratios stay positive for every *Constant* specification, between 0.35 and 0.48: nothing demonstrates a net loss.

Table 8: Pure News Net-of-Cost Performance

This table reports six strategies of the results sections under transaction costs as described in the text. The *Baseline* and *Headline-only* inputs are those of Section 9.9. Each strategy is (ex post) standardized to 10% annual volatility, positions and traded notional together. Alphas are annualized in percent, against the market factor on the broad universe and against the Nasdaq-100 benchmark (QQQ, in excess of the risk-free rate). *t*-statistics (HC3) are reported in parentheses; $n = 143$ monthly observations. Net figures assume a 10 basis point one-way cost per dollar traded. ***, **, and * denote significance at the 1%, 5%, and 10% levels.

			Gross		Net at 10 bps	
	Input	Turnover	SR	α (t)	SR	α (t)
<i>Panel A: Broad Universe</i>						
MSRR <i>Constant</i>	Baseline	0.79	0.79	8.6*** (2.94)	0.35	4.2 (1.46)
	Headline-only	0.79	1.01	10.8*** (3.92)	0.48	5.5** (2.03)
MSRR <i>JKP</i>	Baseline	0.82	0.59	5.8** (2.01)	−0.14	−1.4 (−0.50)
	Headline-only	0.82	0.91	8.2*** (2.86)	−0.08	−1.7 (−0.61)
Pooled Sharpe <i>Constant</i>	Baseline	0.82	0.75	9.1*** (3.09)	0.38	5.4* (1.83)
<i>Panel B: Nasdaq-100</i>						
Pooled Sharpe <i>Constant</i>	Baseline	0.74	0.73	5.3* (1.87)	0.39	2.0 (0.69)

Net alphas separate the two universes. On the Nasdaq-100, the pooled sort adds nothing to its QQQ benchmark that is distinguishable from zero. On the broad universe, the *Headline-only* MSRR retains an alpha against the market factor significant at the 5% level, 5.5% with $t = 2.03$, where its *Baseline* counterpart does not, and the pooled sort holds at the 10% level.²²

DKPT subject their news anomaly to the same exercise and conclude that it “is not an artifact of unrealistic trading costs”. On our corpus, the premise of that anomaly fails upstream of any cost consideration: performance does not increase with residualization depth and even decreases at the deepest rungs. Costs then close the case for the portfolio that carries the claim.

Gross of costs, the signal is there: on the broad universe, every specification carries a significant

²²Against the Fama–French six factors and against the market with the 13 theme portfolios the same net alpha is no longer distinguishable from zero.

alpha against the market. The headline-only input strengthens it further, and even rehabilitates it against the strictest benchmarks, where the baseline was indistinguishable from zero. Net of costs, capture is harder: turnover consumes most of what the signal earns, and in the end a single strategy clears the 5% threshold against its benchmark, the *Constant* specification on headlines alone. On the Nasdaq-100, the alternative selection rules restore positive Sharpe ratios where MSE selection and the MSRR deliver none, but this is not enough: even the most favorably selected configuration falls short of significance at the 5% level against the QQQ benchmark before any cost is charged.

10 Robustness

10.1 Article-to-Month Aggregation

The literature documents three conventions for aggregating article embeddings into a stock-month signal, and they embody different economic bets.

First, our canonical convention averages articles within each news day, then averages across the month's news days, following the released code of [KMSX](#). Each news day carries equal weight: the monthly vector measures the average content of the month's news, regardless of how intensely each day was covered. Same-day repetition is averaged into a single day-mean, and an event that dominates several days of reporting weighs in proportion to those days, not to its article count. Second, the article-mean variant, as described in [DKPT](#), weights all articles of the month equally: the signal follows article volume, so heavily covered days and multi-day events count for proportionally more, potentially tilting the monthly vector toward burst content such as earnings coverage. Finally, a third convention appears in the printed equations of [KMSX](#), the one their authors confirmed to be a typo (Section 5): it divides the sum of the daily means by the number of trading days in the month rather than by the number of days with news. Typo or not, the formula encodes a meaningful economic bet: the monthly vector is scaled by the fraction of trading days on which the stock was covered, so that the vector of a sparsely covered stock is shrunk toward zero, while that of a stock covered on nearly every trading day is left almost unchanged.

Table 9 reports the ladder under the three conventions on identical stock-month cells, keeping everything else fixed, for the *Baseline* and the *Headline-only* inputs. Two results stand out, and both hold under both inputs.

Table 9: Pure News Sharpe Ratios Across Embedding Aggregation Conventions

This table shows the annualized Sharpe ratios of the pure news portfolios on the broad universe under the three monthly aggregation conventions (Day-mean / Article-mean / Trading-day) described in the text, for the *Baseline* and *Headline-only* inputs of Section 9.9, where [†] marks the aggregation convention used throughout the thesis.

	Baseline			Headline-only		
	Day-mean [†]	Article-mean	Trading-day	Day-mean [†]	Article-mean	Trading-day
<i>Constant</i>	0.79	0.85	−0.18	1.01	1.03	−0.22
<i>CAPM</i>	0.82	0.88	−0.17	1.05	1.07	−0.21
<i>FF3</i>	0.78	0.85	0.36	1.16	1.16	0.30
<i>FF6</i>	0.70	0.77	0.26	1.08	1.09	0.23
<i>JKP13</i>	0.68	0.75	0.29	1.06	1.06	0.25
<i>JKP</i>	0.59	0.64	0.10	0.91	0.91	0.09

First, the choice between day-mean and article-mean is immaterial: the article-mean ladder tracks the day-mean closely, sitting slightly above it at every residualized level for the *Baseline* input and matching it under *Headline-only*. The results are robust to this choice. Second, the trading-day convention destroys the signal. The pure news ladder collapses: the two shallowest residualization levels turn negative and the deeper ones retain less than half their day-mean Sharpe ratio. Scaling the monthly embedding by coverage frequency, and hence the position itself, is not a benign renormalization: whatever predictive content it may carry, encoding it this way is counterproductive.

The comparison across inputs adds one reading: the dominance of the *Headline-only* input over the *Baseline* does not depend on the choice between the two valid conventions.

10.2 Point-in-Time Models Comparison

Every result so far rests on news embedded by a single extractor, PIT-4B-FT. This section asks whether the conclusions depend on that choice by re-running the pipeline under the three alternative point-in-time models available: the ChronoGPT variants of He et al. (2025), base and instruct, and the base PIT-4B of KMSX.

Table 10 reports the results. Four readings follow. The first concerns the central negative result of this thesis. No model shows a monotone rise along the ladder: performance peaks at an intermediate rung and declines thereafter, and in seven of the eight cases residualization on the full characteristic set is the weakest of the six. That the signal weakens at the deepest residualization levels is therefore not an artifact of the extractor, nor of the input it reads. The

Table 10: Pure News Sharpe Ratios Across Point-in-Time Models

This table shows the annualized Sharpe ratios of the pure news portfolios on the broad universe, with the news embedded by each of the four point-in-time models (ChronoGPT / ChronoGPT-Instruct / PIT-4B / PIT-4B-FT), for the *Baseline* and *Headline-only* inputs of Section 9.9, where [†] marks the model used throughout the thesis.

	Baseline				Headline-only			
	ChronoGPT	ChronoGPT-Instruct	PIT-4B	PIT-4B-FT [†]	ChronoGPT	ChronoGPT-Instruct	PIT-4B	PIT-4B-FT [†]
<i>Constant</i>	0.64	0.55	0.93	0.79	0.65	0.67	0.88	1.01
<i>CAPM</i>	0.68	0.59	0.97	0.82	0.67	0.68	0.90	1.05
<i>FF3</i>	0.69	0.66	1.02	0.78	0.74	0.78	1.00	1.16
<i>FF6</i>	0.66	0.65	1.01	0.70	0.69	0.77	0.95	1.08
<i>JKP13</i>	0.59	0.59	0.98	0.68	0.63	0.72	0.92	1.06
<i>JKP</i>	0.41	0.51	0.63	0.59	0.55	0.70	0.75	0.91

second replicates [KMSX](#). Scale dominates: both 4B variants beat both ChronoGPT variants at every rung and under both inputs.

The third qualifies their reading of fine-tuning. On the *Baseline* input, PIT-4B-FT does not outperform PIT-4B at any rung of the ladder; on the *Headline-only* input the ranking reverses, and the fine-tuned model wins at every rung. [KMSX](#) suggest that fine-tuning “might improve robustness and generalization of the extracted signals across the cross-section”; on this corpus, this result depends on the input the model reads. Finally, the table allows a reading across inputs: the gain of the *Headline-only* input over the *Baseline*, documented in Section 9.9 for PIT-4B-FT, does not carry over to all point-in-time models tested.

10.3 The Ensemble Volatility-Scaling Convention

Each value of the MSRR shrinkage grid defines a strategy, and the grid is equally weighted after each strategy has been rescaled to a 10% target volatility (Section 8), following [KMSX](#). Their released code estimates each strategy’s volatility over the full prediction sample, and we follow it throughout the thesis: under this ex-post scaling, the factor applied to a given strategy incorporates information from months that postdate the returns being scaled. This ex-post estimation departs from the point-in-time discipline maintained everywhere else in the pipeline, so we test whether it matters with a strictly point-in-time (PIT) variant. Each strategy’s rescaling at month t uses the volatility of its realized returns through month $t - 1$ only: the first out-of-sample month (January 2014) is thus scaled by the volatility estimated over the entire preceding warm-up window (August 2010 to December 2013), and the estimation window expands by one month at

each subsequent step. The one-month lag ensures that no contemporaneous information enters the scaling. As the window grows, this expanding estimate converges to the full-sample value, so the two conventions increasingly coincide toward the end of the sample.

Table 11: Pure News Sharpe Ratios Across Volatility-Scaling Conventions

This table shows the annualized Sharpe ratios of the pure news portfolios on the broad universe under the two conventions used to rescale each MSRR shrinkage strategy to a 10% annualized target volatility before averaging, for the *Baseline* and *Headline-only* inputs of Section 9.9: Ex-Post estimates each strategy’s volatility over the full prediction sample, PIT over an expanding, one-month-lagged window; [†] marks the convention used throughout the thesis.

	Baseline		Headline-only	
	Ex-Post [†]	PIT	Ex-Post [†]	PIT
<i>Constant</i>	0.79	0.86	1.01	0.98
<i>CAPM</i>	0.82	0.90	1.05	1.03
<i>FF3</i>	0.78	0.89	1.16	1.15
<i>FF6</i>	0.70	0.79	1.08	1.08
<i>JKP13</i>	0.68	0.75	1.06	1.05
<i>JKP</i>	0.59	0.69	0.91	0.93

Table 11 reports the outcome for the *Baseline* and the *Headline-only* inputs. The ex-post convention buys no performance: on the *Baseline* input, the expanding point-in-time alternative delivers marginally higher Sharpe ratios at every rung of the ladder, and on the *Headline-only* input the two conventions coincide to within 0.05. Our results therefore owe nothing to this departure from the point-in-time discipline.

10.4 Performance by Subperiod

We ask whether the results are concentrated in part of the out-of-sample window. The sample splits into two near-halves at the end of 2019, 72 formation months before and 71 after, coinciding with the corpus regime change, the editorial shift documented in Section 6.

Table 12 reports the Sharpe ratios of the pure news portfolios in each half and over the full window, for both inputs at every residualization level. Over 2014–2019, the two inputs behave alike at every level. Performance declines markedly along the residualization ladder, from 1.18 and 1.12 at *Constant* to 0.35 and 0.45 at the full *JKP* set. The two inputs truly diverge only in the second half: over 2020–2025, the *Headline-only* portfolios dominate at every residualization level, with Sharpe ratios ranging from 1.08 to 1.41 across the ladder, against 0.71 to 0.87 for the *Baseline*, whose profile is mostly flat where the *Headline-only* one peaks at intermediate levels; under either input, the *JKP* set is no longer the weakest. Throughout, the *Constant* specification

Table 12: Pure News Sharpe Ratios by Subperiod

This table shows the annualized Sharpe ratios of the pure news portfolios on the broad universe over the two halves of the out-of-sample window and over the full window, for the *Baseline* and *Headline-only* inputs of Section 9.9. Windows are defined on the formation month, 2014–2019 covering 72 formations and 2020–2025 the remaining 71.

	Baseline			Headline-only		
	2014–2019	2020–2025	Full	2014–2019	2020–2025	Full
<i>Constant</i>	1.18	0.81	0.79	1.12	1.08	1.01
<i>CAPM</i>	1.09	0.87	0.82	1.07	1.16	1.05
<i>FF3</i>	0.99	0.83	0.78	0.92	1.41	1.16
<i>FF6</i>	0.98	0.71	0.70	0.92	1.29	1.08
<i>JKP13</i>	0.81	0.76	0.68	0.80	1.33	1.06
<i>JKP</i>	0.35	0.79	0.59	0.45	1.23	0.91

holds Sharpe ratios between 0.81 and 1.18 across halves and inputs.

The improvement of the *Headline-only* input thus coincides with the editorial shift, but it does not extend to the *Baseline*, which performs below its first-half levels everywhere but at the *JKP* set. If the headline gain were only a gain in coverage due to the shift, it would show up in both inputs, since the news items are the same. Pinpointing the exact driver is difficult: the editorial shift affects the breadth of the corpus, but potentially also the nature of its content, part of the production becoming automated. More generally, the differences in performance and in patterns between the two halves cannot be attributed to the editorial shift alone, the market environment being different as well, from the COVID episode and its rebound to the AI boom.

10.5 Rolling Scheme and Lookahead Bias

Our main results rely on the rolling-checkpoint scheme of Section 5, which enforces point-in-time validity by design and serves as our clean benchmark. To assess how relaxing it affects performance we embed the entire sample with a single fixed model vintage, varied from the earliest (2013) to the latest (2024) checkpoint, and compare the resulting Sharpe ratios with those of the rolling benchmark. We adopt the design of He et al. (2025) and run it at each level of the residualization ladder. For computational reasons, this exercise is run on the *Baseline* input only.

Figure 12 plots, for four residualization specifications (*Constant*, *CAPM*, *JKP13*, and *JKP*), the Sharpe ratio obtained when the whole corpus is embedded with a single fixed vintage (solid), against the rolling-checkpoint benchmark (dashed). *FF3* and *FF6* are omitted for space; their vintage profiles track that of *JKP13* almost exactly. Since a later vintage applied to the whole history yields embeddings from a model that was trained on more future information, lookahead

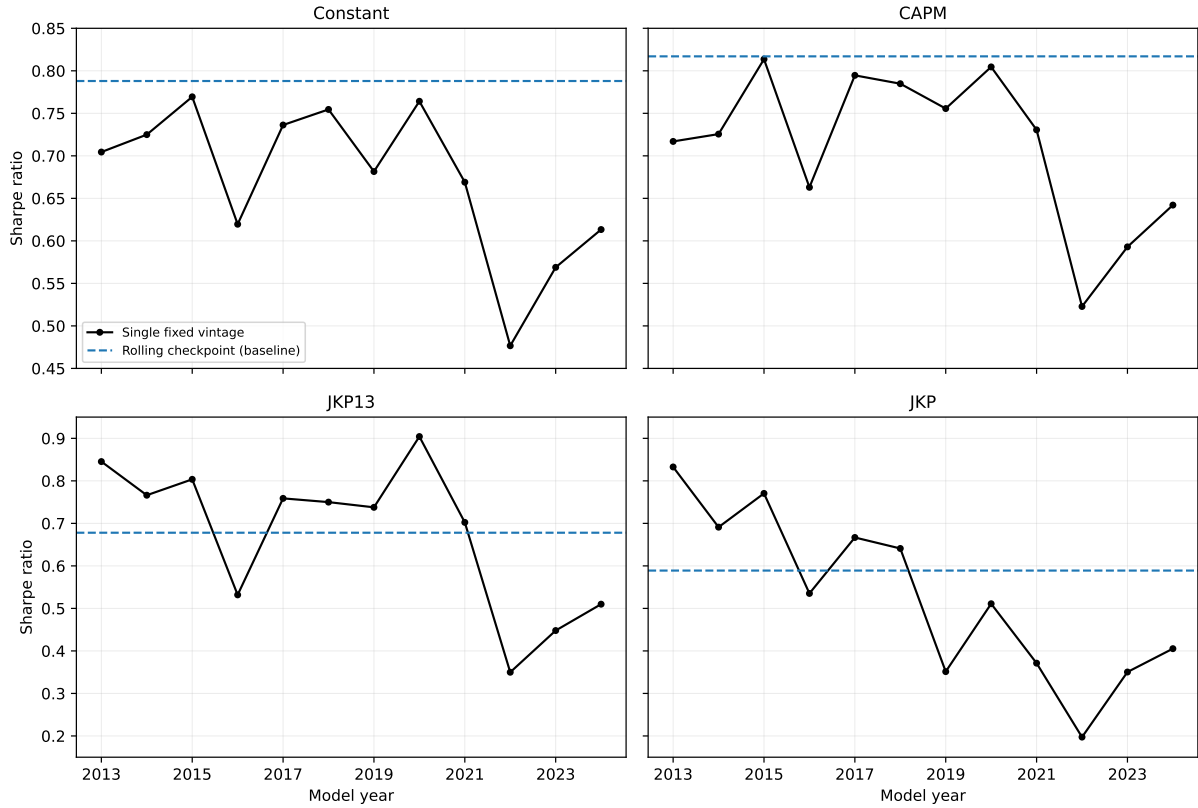


Figure 12: Pure News Sharpe Ratios Across Vintage Checkpoints

Note. This figure shows the annualized pure news Sharpe ratios on the broad universe under single fixed vintage checkpoints (solid line) and under our baseline rolling-checkpoint scheme (dashed line), for four residualization levels. In the first, the entire corpus is embedded with one PIT-4B-FT checkpoint, given by the model year on the horizontal axis; in the second, each article is embedded with the most recent checkpoint that strictly predates its publication year (Section 5). The vertical scale is common within each row and differs between rows.

bias, were it a driver, would show up as the Sharpe ratio rising with the vintage year.

The figure shows no such rise. Under *Constant*, *CAPM*, and *JKP13* the Sharpe ratio fluctuates without trend through the 2021 checkpoint, and then deteriorates at the most recent vintages, while under *JKP* it drifts downward across the whole sweep. Part of the variation, including the deterioration at the most recent vintages, may come from the training data itself: each vintage extends the previous one with an additional year of FineWeb data, whose composition need not be comparable across years. What matters here is the absence of a rise. The 2024 checkpoint, whose training data run through the end of 2024 and therefore overlap with nearly the entire evaluation window, carries the greatest potential lookahead exposure. It lies below the earliest checkpoints and below the rolling benchmark, and its 2022 and 2023 neighbours fare no better. At the other end, the 2013 checkpoint, the least exposed of the twelve, is the strongest under *JKP* and second only to 2020 under *JKP13*.

The literature runs a similar comparison, between the rolling scheme and a fixed checkpoint

whose training spans its entire sample. [KMSX](#) use their latest checkpoint, dated December 2024, and find that their rolling scheme performs even better, so that enforcing the point-in-time constraint costs nothing. [DKPT](#) use a checkpoint trained through 2022, the last year of their sample, and find the two schemes essentially indistinguishable, the fixed checkpoint in fact marginally behind. Our result agrees with theirs, on a claim that needs scoping: enforcing point-in-time integrity costs no performance relative to the checkpoints most exposed to the sample’s future. The sweep does, however, put the one-vintage version of this test in question. Its verdict depends on which checkpoint is chosen to stand for the contaminated model: under *JKP13*, choosing the 2020 checkpoint would have placed it ahead of the rolling scheme, and choosing the 2022 one behind. A verdict read off a single checkpoint is therefore specific to the checkpoint chosen, where a profile across twelve is not.

These observations suggest that lookahead bias is a minor concern in our setting, the inference drawn by [He et al. \(2025\)](#) from an analogous sweep and, in its single-checkpoint form, by both papers above. This inference, however, warrants qualification. Moving to a later checkpoint may change two things at once: the checkpoint has been trained on more of the article’s future (the lookahead channel), but it also reads the article through the semantics of a different period (the alignment channel). The two effects may have opposite impacts on performance and cannot be varied separately: the vintage profile identifies only their net, and a material lookahead gain offset by an alignment penalty could produce the very profile we observe. [He et al. \(2025\)](#) invoke the alignment channel as a potential explanation for why their rolling scheme lies above most fixed vintages, but do not carry it back to the lookahead inference itself. The offsetting scenario is nonetheless unlikely: as [DKPT](#) note, the downstream task (return prediction) is decoupled from the model’s training objective (token prediction); the lookahead bias documented in the literature concerns text generation, which derives directly from that objective, and it has not been shown that embedding-based applications inherit it.

The rolling scheme, by contrast, needs no trade-off between the two: it is free of lookahead by construction while keeping each article’s checkpoint close to its own period. It lies above all twelve checkpoints under *Constant* and *CAPM*, and in the middle of them from *FF3* to *JKP*, so its advantage exists only at the two shallowest residualization levels. Why is unclear: if temporal alignment is what lifts it at those rungs, residualization may strip out the component alignment acts on. We leave the question open.

11 Conclusion

This thesis asked whether accessible news predicts returns, and whether purging news of its predictable content strengthens the signal, as it does on the premium newswires of the literature. We replicated the pipeline of [KMSX](#), from chronologically consistent embeddings to monthly news portfolios, and the pure news design of [DKPT](#), on 1.6 million Benzinga articles over two point-in-time universes, the Nasdaq-100 and a broad universe averaging over two thousand firms a month, with returns evaluated out of sample from 2014 to 2025.

The pure news anomaly. On premium news, the signal strengthens as characteristics are stripped from the embeddings and, at the full set, delivers what [DKPT](#) name the pure news anomaly. That anomaly does not extend to our accessible corpus. On the broad universe, the only one wide enough to carry the comparison, the predictable content is there: firm characteristics explain a share of our demeaned embeddings similar to what they report. The purge, however, does not pay. The one consistent gain of the residualization ladder is the cross-sectional demean; beyond it, residualizing adds little at best and costs performance at the deeper sets, where the full *JKP* set is systematically the weakest, with a Sharpe ratio of 0.59 against 0.79 at *Constant*. Theirs rises through the full *JKP* set, saturating at worst but never declining. We show that the decline does not depend on which characteristics the ladder uses: random subsets of k characteristics trace the same downward path as k grows, where theirs rises concavely.

The factor exposures tell the same story. The purge does its mechanical work on both corpora, the share of portfolio variance explained by the *JKP* themes falling from 30% to 10% on ours as from 40% to 11% on theirs; but the decoupling is compensated only on premium news. Their theme alpha grows with the purge, from 14.6% to 29.4% per year at 10% volatility; ours sits between 3.9% and 4.9%, indistinguishable from zero at every depth. The signal survives against the market factor, from 8.6% ($t = 2.94$) to 5.8% ($t = 2.01$) across the ladder, and even there the tiers diverge: read from their published figure, their market alpha runs from about 15% at the demean to 30% at the full set, rising where ours declines.

Substituting accessible news for the premium newswires has a price, which we quantify as the *Premium Gap*. Under configurations closely aligned with each reference paper, the gap is positive wherever a benchmark exists: close to 60% of the premium Sharpe ratio is retained at the shallowest residualization, a quarter to a little more than a tenth at the deepest, where the gap is corroborated by two distinct premium newswires, Dow Jones and Reuters.

The question left is why. [DKPT](#) read pure news as the content of the text that is unexpected relative to the residualization set, and its performance as the market's delayed reaction to that content. On our corpus, that interpretation finds little to explain: deeper purges uncover no additional performance, so either the component of news orthogonal to characteristics carries less of the information that re-values assets than its premium counterpart, or the market absorbs most of it without delay. The design cannot separate the two readings, and we leave the question open; under either, the anomaly that motivates the deepest residualization belongs to the premium tier.

The accessible news signal. The signal that survives against the market factor is largely robust to a change of estimator, and confined to the broad universe. There, replacing the MSRR estimator with a pooled ridge, which predicts next-month returns from the embeddings and sorts on the forecasts, changes little down to *JKP13*: performance stays close, at times slightly higher, and does not systematically increase with deeper residualization. The full *JKP* set is the exception: the sort collapses when the ridge penalty is selected on prediction error, and recovers when it is selected on the quality of the ranking instead. On the Nasdaq-100, neither estimator delivers anything. The same ranking-based selection restores positive Sharpe ratio point estimates, strongest at the lightest residualization, but the best of them stays at most marginally significant against the QQQ benchmark.

The input, by contrast, matters. On the broad universe, the strongest signal lives in the news headline and the body dilutes it. Under the headline input, the alphas are higher against every benchmark we use: against the CAPM, from 10.8% ($t = 3.92$) at *Constant* to 8.2% ($t = 2.86$) at *JKP*; against the Fama–French six factors, from 8.1% ($t = 2.90$) to 8.0% ($t = 2.59$); and against the theme portfolios, from 7.3% ($t = 2.45$) to 7.1% ($t = 2.16$), the one model against which the baseline input earns nothing distinguishable from zero.

Net of a realistic transaction cost, little survives. The *JKP* pure news portfolio, the one that carries the literature's claim, turns negative. The *Constant* specifications keep positive net Sharpe ratios on both universes, between 0.35 and 0.48. However, statistical significance at the 5% level remains on the headline input alone, on the broad universe for the *Constant* specification, with a market alpha of 5.5% ($t = 2.03$). This has a practical corollary: the best-performing input, and the only one whose significance against its benchmark survives transaction costs, is also an order of magnitude cheaper to embed. The saving is directly relevant to the audience an accessible provider might serve.

Robustness. We finally ask how much these results owe to our implementation choices, and whether they are concentrated in part of the sample. Every check runs on the broad universe with the MSRR estimator, under both the baseline and the headline-only inputs. First, the choice between the two valid article-to-month aggregation schemes of our reference literature, averaging by news day or by article, is immaterial. Second, the choice of embedding extractor is not innocuous and moves the Sharpe ratio levels. Scale dominates, as in [KMSX](#): the PIT-4B variants beat the ChronoGPT variants uniformly. Within the PIT-4B pair, the fine-tuned variant costs performance when it reads the baseline input, and wins at every level when it reads the headlines alone; the headline advantage is itself extractor-dependent and does not carry over to the base variant. Third, the ex-post volatility scaling of the ensemble shrinkage strategies, inherited from [KMSX](#), does not inflate our performance: an expanding point-in-time alternative delivers similar results. Across every variant of these three exercises, the shape of the ladder is the same: performance peaks at an intermediate residualization level and declines thereafter, with the *JKP* specification almost always the weakest. Hence, this pattern is not an artifact of the aggregation scheme, the embedding extractor, or the text the model reads.

Splitting the out-of-sample window into two halves qualifies these results. In the first half, the two inputs are indistinguishable and the ladder declines, most of it at the full *JKP* set. In the second, the headline-only input dominates at every level, its profile peaking at intermediate purges, and the *JKP* set is no longer the weakest. The *Constant* specification retains Sharpe ratios between 0.81 and 1.18 across halves and inputs.

Our rolling-checkpoint scheme rules out training leakage by construction; rerunning the pipeline on embeddings extracted from each of twelve fixed vintage checkpoints shows no rise of the Sharpe ratio with the vintage year, suggesting that this bias is a minor concern in our setting. The sweep also reveals what a single-checkpoint comparison can hide, the verdict depending on the checkpoint chosen to stand for the contaminated model.

Limitations. First, the comparisons with the literature are not equally tight. The main analysis of [DKPT](#) is conducted on embeddings extracted with E5-Mistral-7B, under their own conventions, mean pooling included; the $SR(k)$ profile, the breadth analysis, and the factor exposures are reported for that extractor only, unlike the reference values we use for the Premium Gap. No tighter comparison is therefore available for these objects, and we mainly compare patterns rather than levels. Against [KMSX](#), the protocol is replicated exactly. Second, the non-replication of the

pure news anomaly is a full-sample statement, and its two components do not carry equally to the subperiods: performance builds toward the full characteristic set in neither half, under either input, but the decline documented over the full sample is concentrated in the first half and does not extend to the second. Third, everything rests on a single provider. The corpus, the ticker tags on which the attribution of articles to stocks relies, and the editorial conventions are all Benzinga's, and we cannot separate what is specific to this feed from what is generic to accessible news. The Premium Gap we document has the same bound: it prices a package, not its channels. Fourth, the implementable version of the signal remains fragile. Net of costs, significance at the 5% level is confined to the headline input and the lightest residualization level, with a t -statistic barely clearing the threshold; nothing deeper survives.

Further research. Several directions follow. First, replaying the pipeline on other accessible providers, or on a combination of them, would show how much of our results is specific to one feed and would reduce the dependency described above. Second, the signal is real but its standalone tradability is limited; a natural next step is to exploit it in combination, merging it with strategies built on other inputs, fundamental data for instance, or using it as a risk layer on top of them. Third, the exercise is confined to US equities; whether accessible news carries a signal in other markets is an open question. Finally, other paths of the news-based literature remain unexplored on accessible news, such as the daily design of [CKX](#) or supervised fine-tuning methods applied to point-in-time language models.

References

- Alpha Vantage. Market news & sentiment. Alpha Vantage API Documentation, <https://www.alphavantage.co/documentation/#news-sentiment>. Accessed April 2026.
- Dogu Araci. FinBERT: Financial sentiment analysis with pre-trained language models. *arXiv preprint arXiv:1908.10063*, 2019.
- Clifford S Asness, Andrea Frazzini, and Lasse Heje Pedersen. Quality minus junk. *Review of Accounting Studies*, 24(1):34–112, 2019.
- Rolf W Banz. The relationship between return and market value of common stocks. *Journal of Financial Economics*, 9(1):3–18, 1981.
- Johan Bollen, Huina Mao, and Xiaojun Zeng. Twitter mood predicts the stock market. *Journal of Computational Science*, 2(1):1–8, 2011.
- Leland Bybee, Bryan Kelly, Asaf Manela, and Dacheng Xiu. Business news and business cycles. *The Journal of Finance*, 79(5):3105–3147, 2024.
- Charles W Calomiris and Harry Mamaysky. How news and its context drive risk and returns around the world. *Journal of Financial Economics*, 133(2):299–336, 2019.
- Yifei Chen, Bryan T Kelly, and Dacheng Xiu. Expected returns and large language models. *Available at SSRN 4416687*, 2022.
- Michael J Cooper, Huseyin Gulen, and Michael J Schill. Asset growth and the cross-section of stock returns. *The Journal of Finance*, 63(4):1609–1651, 2008.
- Antoine Didisheim, Shikun Ke, Bryan Kelly, and Semyon Malamud. APT or “AIPT”? the surprising dominance of large factor models. Technical report, National Bureau of Economic Research, 2024.
- Antoine Didisheim, Bryan T Kelly, Mohammad Pourmohammadi, and Hanqing Tian. The inefficient pricing of news. Technical report, National Bureau of Economic Research, 2026. Version of 20 May 2026.
- Zihan Dong, Xinyu Fan, and Zhiyuan Peng. FNSPID: A comprehensive financial news dataset in time series. In *Proceedings of the 30th ACM SIGKDD Conference on Knowledge Discovery and Data Mining*, pages 4918–4927, 2024.

- Joseph Engelberg, Asaf Manela, William Mullins, and Luka Vulicevic. Entity neutering. *Available at SSRN 5182756*, 2026.
- EODHD. Financial news feed and stock news sentiment data api. EODHD APIs Documentation, <https://eodhd.com/financial-apis/stock-market-financial-news-api>. Accessed April 2026.
- Eugene F Fama and Kenneth R French. A five-factor asset pricing model. *Journal of Financial Economics*, 116(1):1–22, 2015.
- Eugene F Fama and James D MacBeth. Risk, return, and equilibrium: Empirical tests. *Journal of Political Economy*, 81(3):607–636, 1973.
- Andrea Frazzini, Ronen Israel, and Tobias J Moskowitz. Trading costs. *Available at SSRN 3229719*, 2018.
- Matthew Gentzkow, Bryan Kelly, and Matt Taddy. Text as data. *Journal of Economic Literature*, 57(3):535–574, 2019.
- Paul Glasserman and Caden Lin. Assessing look-ahead bias in stock return predictions generated by gpt sentiment analysis. *arXiv preprint arXiv:2309.17322*, 2023.
- Wenjun Gu, Yihao Zhong, Shizun Li, Changsong Wei, Liting Dong, Zhuoyue Wang, and Chao Yan. Predicting stock prices with FinBERT-LSTM: Integrating news sentiment analysis. In *Proceedings of the 2024 8th International Conference on Cloud and Big Data Computing*, pages 67–72, 2024.
- Tian Guo and Emmanuel Hauptmann. Fine-tuning large language models for stock return prediction using newsflow. In *Proceedings of the 2024 Conference on Empirical Methods in Natural Language Processing: Industry Track*, pages 1028–1045, 2024.
- Campbell R Harvey and Akhtar Siddique. Conditional skewness in asset pricing tests. *The Journal of Finance*, 55(3):1263–1295, 2000.
- Songrun He, Linying Lv, Asaf Manela, and Jimmy Wu. Chronologically consistent large language models. *arXiv preprint arXiv:2502.21206*, 2025.
- Narasimhan Jegadeesh. Evidence of predictable behavior of security returns. *The Journal of Finance*, 45(3):881–898, 1990.

- Narasimhan Jegadeesh and Sheridan Titman. Returns to buying winners and selling losers: Implications for stock market efficiency. *The Journal of Finance*, 48(1):65–91, 1993.
- Theis Ingerslev Jensen, Bryan Kelly, and Lasse Heje Pedersen. Is there a replication crisis in finance? *The Journal of Finance*, 78(5):2465–2518, 2023.
- Guohua Jiang, Charles MC Lee, and Yi Zhang. Information uncertainty and expected returns. *Review of Accounting Studies*, 10(2-3):185–221, 2005.
- Zheng Tracy Ke, Bryan T. Kelly, and Dacheng Xiu. Predicting returns with text data. NBER Working Paper 26186, National Bureau of Economic Research, 2019.
- Bryan Kelly and Dacheng Xiu. Financial machine learning. *Foundations and Trends® in Finance*, 13(3-4):205–363, 2023.
- Bryan T Kelly, Semyon Malamud, Johannes Schwab, and Teng Andrea Xu. Scaling point-in-time language models. *Swiss Finance Institute Research Paper*, (26-37), 2026. Version of 30 April 2026.
- Kemal Kirtac and Guido Germano. Sentiment trading with large language models. *Finance Research Letters*, 62:105227, 2024.
- Yaxuan Kong, Hoyoung Lee, Yoontae Hwang, Alejandro Lopez-Lira, Bradford Levy, Dhagash Mehta, Qingsong Wen, Chanyeol Choi, Yongjae Lee, and Stefan Zohren. Evaluating llms in finance requires explicit bias consideration. *arXiv preprint arXiv:2602.14233*, 2026.
- Alejandro Lopez-Lira and Yuehua Tang. Can ChatGPT forecast stock price movements? return predictability and large language models. *arXiv preprint arXiv:2304.07619*, 2023.
- Alejandro Lopez-Lira, Yuehua Tang, and Mingyin Zhu. The memorization problem: Can we trust LLMs’ economic forecasts? *arXiv preprint arXiv:2504.14765*, 2025.
- Tim Loughran and Bill McDonald. When is a liability not a liability? textual analysis, dictionaries, and 10-ks. *The Journal of Finance*, 66(1):35–65, 2011.
- Linying Lv. The value of information from sell-side analysts. *arXiv preprint arXiv:2411.13813*, 2024.
- Asaf Manela and Alan Moreira. News implied volatility and disaster concerns. *Journal of Financial Economics*, 123(1):137–162, 2017.

- Massive. Real-time Benzinga News. Massive API Documentation, <https://massive.com/docs/rest/partners/benzinga/news>. Accessed April 2026.
- S McKay Price, James S Doran, David R Peterson, and Barbara A Bliss. Earnings conference calls and stock returns: The incremental informativeness of textual tone. *Journal of banking & finance*, 36(4):992–1011, 2012.
- Scott A Richardson, Richard G Sloan, Mark T Soliman, and Irem Tuna. Accrual reliability, earnings persistence and stock prices. *Journal of Accounting and Economics*, 39(3):437–485, 2005.
- Barr Rosenberg, Kenneth Reid, and Ronald Lanstein. Persuasive evidence of market inefficiency. *The Journal of Portfolio Management*, 11(3):9–16, 1985.
- Suproteem K Sarkar. Storieslm: A family of language models with time-indexed training data. *Available at SSRN 4881024*, 2024.
- Suproteem K. Sarkar and Keyon Vafa. Lookahead bias in pretrained language models. *SSRN working paper*, 2024.
- Richard G Sloan. Do stock prices fully reflect information in accruals and cash flows about future earnings? *The Accounting Review*, 71(3):289–315, 1996.
- Paul C Tetlock. Giving content to investor sentiment: The role of media in the stock market. *The Journal of finance*, 62(3):1139–1168, 2007.
- Paul C. Tetlock. All the news that’s fit to reprint: Do investors react to stale information? *The Review of Financial Studies*, 24(5):1481–1512, 2011.
- Paul C. Tetlock, Maytal Saar-Tsechansky, and Sofus Macskassy. More than words: Quantifying language to measure firms’ fundamentals. *The Journal of Finance*, 63(3):1437–1467, 2008.
- Jacob Thomas and Frank X Zhang. Tax expense momentum. *Journal of Accounting Research*, 49(3):791–821, 2011.
- Yutong Yan, Raphael Tang, Zhenyu Gao, Wenxi Jiang, and Yao Lu. Datedgpt: Preventing lookahead bias in large language models with time-aware pretraining. *arXiv preprint arXiv:2603.11838*, 2026.

Appendices

A Padding and the Reproducibility of News Embeddings

The embeddings used throughout this thesis are extracted with the procedure described in Section 5, which builds on the code released alongside [KMSX](#). This section documents a reproducibility issue we found in that code, its diagnosis, and its resolution.

The symptom. The issue surfaced when we extended the pipeline from the Nasdaq-100 to the broad universe. On the articles shared by the two runs, identical texts produced embeddings that differed beyond numerical noise. The only difference between the runs was the composition of the batches in which the articles were embedded, pointing to batch composition as the culprit.

The extraction code. The extraction routine in the GitHub repository (`embed_articles` in `embeddings/4b/main.py`; Listing 1) reads the hidden state of each sequence’s last token and pads batches on the left, so that the last real token sits at the final position (“left-padded so last token is at -1”). The model itself, however, applies causal attention only (`is_causal=True` in `models/GPT.py`): its attention layer takes no mask argument, so padding positions cannot be masked. This left-padding also stands alone in the repository: the benchmark evaluation of PIT-4B right-pads (`pad_sequence` in `models/PIT.py`), the evaluation code of a related causal model annotates its batching “right-pad is safe for causal LMs” (`eval/eval_chrono.py`), and the financial-application code loads all embedding sets under `-right` identifiers (`PIT-4B-FT-right`).

Listing 1: Last-token extraction with left-padding.

```
padded = [[0] * (max_len - len(t)) + t for t in token_ids]
...
# (batch, seq_len, n_embd) - left-padded so last token is at -1
batch_emb = captured["hidden"][:, -1, :]
```

Note: Excerpt from the function `embed_articles` in `embeddings/4b/main.py`, in the released repository at commit `1e17a9f`. Each token sequence is prefixed with zeros up to the batch maximum, so the last real token sits at the final position, where the embedding is read.

Why padding side matters. The causal mask hides what lies to a token’s *right*; a padding mask, absent here, would hide the padding itself. Under right-padding, the padding sits to the right of the last real token and is excluded by the causal mask, which is why right-padding is safe for a causal model. Under left-padding, it sits inside the last token’s causal window and is attended to:

since each batch is padded dynamically to its longest article, the extraction is batch-dependent, so the embedding is not a function of the article alone and the same code returns different results whenever the batching changes.

The resolution. We reported the issue to the authors, who confirmed it: the released script was an earlier version, and the embeddings used in the paper had in fact been produced with right-padding. The released code was thus not the code that generated the paper’s embeddings, and the paper’s results were not reproducible from it. The repository was corrected shortly after (commit 8136b09; Listing 2): `embed_articles` now takes a `padding` argument, pads on the right by default, and reads each article’s last token at its true length rather than at position `-1`.

Listing 2: Corrected extraction with right-padding.

```
if padding == "right":
    padded = [t + [0] * (max_len - len(t)) for t in batch_ids]
...
model(input_ids, full_sequence=(padding == "right"))
hidden = captured["hidden"]
if padding == "right":
    batch_emb = np.stack([hidden[j, lengths[j] - 1, :]
                          for j in range(len(batch_ids))])
```

Note: Excerpt from the updated `embed_articles` (`embeddings/4b/main.py` function, commit 8136b09), now parameterised by padding side. The right-padding branch pads after the real tokens and reads the last real token by its true length, rather than at the fixed position `-1`. Right-padding is the default.

All embeddings in this thesis are extracted under the corrected right-padding convention, and the corrected release now reproduces the paper’s embeddings for any future user of the point-in-time models.

B Residualization Mapping

We residualize the news embeddings against a ladder of factor specifications of increasing richness. Table B.1 maps each specification to the JKP characteristics it controls for; the two paragraphs below explain how that mapping is built and how *JKP13* is selected.

Mapping. [Didisheim et al. \(2026\)](#) describe each rung by the Fama–French factors it controls for, not by the specific JKP characteristic that measures each one: *CAPM* for market beta, *FF3* adds size and book-to-market, and *FF6* adds profitability, investment, and momentum. We recover that characteristic by matching the Fama–French construction, as documented in Kenneth French’s

Table B.1: Residualization Specifications and Their JKP Characteristics

This table shows each residualization specification, the JKP theme(s) it controls for, the characteristic used to measure each theme, and its original reference. Specifications are nested ($Constant \subset CAPM \subset FF3 \subset FF6 \subset JKP13$); “(+)” marks the characteristics each rung adds to the one above it. *Constant* is cross-sectional demeaning (intercept only), while *JKP* is the coverage-filtered set of 130 characteristics and, being defined by coverage rather than by nesting, is not in general a strict superset of *JKP13*. Themes, characteristic names, codes, and original references follow the JKP documentation (Jensen et al., 2023, Table 9).

Specification	Theme	Description	Variable name	Citation
Constant	—	—	—	—
CAPM (+)	Low Risk	Market Beta	beta_60m	Fama and MacBeth (1973)
FF3 (+)	Size	Market Equity	market_equity	Banz (1981)
	Value	Book-to-market equity	be_me	Rosenberg et al. (1985)
FF6 (+)	Profitability	Operating profits-to-book equity	ope_be	Fama and French (2015)
	Investment	Asset Growth	at_gr1	Cooper et al. (2008)
	Momentum	Price momentum t-12 to t-1	ret_12_1	Jegadeesh and Titman (1993)
JKP13 (+)	Accruals	Operating accruals	oaccruals_at	Sloan (1996)
	Short-Term Reversal	Short-term reversal	ret_1_0	Jegadeesh (1990)
	Debt Issuance	Change in financial liabilities	fnl_gr1a	Richardson et al. (2005)
	Low Leverage	Firm age	age	Jiang et al. (2005)
	Seasonality	Coskewness	coskew_21d	Harvey and Siddique (2000)
	Quality	Quality minus Junk: Safety	qmj_safety	Asness et al. (2019)
	Profit Growth	Tax expense surprise	tax_gr1a	Thomas and Zhang (2011)
JKP	all	Full coverage-filtered set	130 characteristics	Jensen et al. (2023)

data library,²³ which states what each factor sorts on, against the JKP documentation (Jensen et al., 2023), which states how each characteristic is built.

Each *FF6* characteristic is thus the JKP counterpart of its sort variable. The correspondence holds up to update cadence and missing-data rules, which leave the variable definitions unchanged: *be_me*, for instance, uses current-month rather than frozen December market equity.

JKP13 selection. *JKP13* uses one representative characteristic per JKP theme. The six *FF6* sort variables represent their own themes. For Accruals and Short-Term Reversal we use the characteristic canonical in the literature: *oaccruals_at* and *ret_1_0*. The remaining five themes are represented by the best-covered characteristic on the Nasdaq-100; this fixes a single *JKP13* composition, which we apply unchanged to the broad universe so that specifications are comparable across universes. For two of them, Debt Issuance and Profit Growth, the best coverage is exactly tied, so the choice is arbitrary; we fix it by a frozen convention (*fnl_gr1a* and *tax_gr1a*) for reproducibility.

²³Accessed April 2026: https://mba.tuck.dartmouth.edu/pages/faculty/ken.french/data_library.html

C Additional Tables

Table C.1: Pure News Portfolio Sorts on the MSRR Weights

This table shows equal-weighted quintile portfolios formed each month by sorting stocks on the per-stock weights of the MSRR strategy, for each residualization specification. Panel A reports the broad universe, Panel B the Nasdaq-100. The first five columns report the annualized average return of each score quintile, in excess of the risk-free rate (%). The remaining columns report the high-minus-low spread: annualized mean (%), volatility (%), CAPM alpha (%) against the market factor in Panel A and against the Nasdaq-100 benchmark (QQQ, in excess of the risk-free rate) in Panel B, its HC3 t -statistic, and the annualized Sharpe ratio. $n = 143$ monthly observations.

Panel A: Broad Universe										
	Excess return by score quintile					High-minus-low spread				
	Low	2	3	4	High	Mean	Vol	α	$t(\alpha)$	SR
<i>Constant</i>	2.6	9.1	9.4	8.7	9.3	6.7	6.7	7.0	3.55	1.00
<i>CAPM</i>	2.5	8.7	9.5	8.8	9.7	7.2	6.6	7.4	3.83	1.10
<i>FF3</i>	3.4	9.0	9.5	9.2	8.0	4.6	5.7	5.2	3.31	0.81
<i>FF6</i>	3.9	9.2	9.2	9.0	7.8	4.0	5.3	4.3	2.86	0.75
<i>JKP13</i>	4.3	9.3	8.8	8.9	7.9	3.6	4.8	3.3	2.44	0.74
<i>JKP</i>	5.9	8.8	8.1	8.3	8.1	2.2	4.1	1.7	1.42	0.54

Panel B: Nasdaq-100										
	Excess return by score quintile					High-minus-low spread				
	Low	2	3	4	High	Mean	Vol	α	$t(\alpha)$	SR
<i>Constant</i>	11.5	14.5	13.2	13.3	13.1	1.6	11.3	-1.3	-0.40	0.14
<i>CAPM</i>	12.9	12.4	14.0	11.8	14.1	1.2	11.6	-1.6	-0.48	0.11
<i>FF3</i>	12.8	12.0	15.6	12.2	12.8	-0.1	11.3	-1.4	-0.43	-0.00
<i>FF6</i>	13.2	13.5	12.6	13.0	13.1	-0.2	10.1	-2.0	-0.67	-0.02
<i>JKP13</i>	12.1	13.6	11.0	15.6	13.1	1.0	10.3	0.5	0.15	0.10

Table C.2: Pure News Portfolio Sorts Under Rank-IC Selection

This table shows equal-weighted quintile portfolios formed each month by sorting stocks on the predicted return of the pooled ridge, with the ridge penalty selected by the Rank-IC rule defined in the text, for each residualization specification. Panel A reports the broad universe, Panel B the Nasdaq-100. The first five columns report the annualized average return of each score quintile, in excess of the risk-free rate (%). The remaining columns report the high-minus-low spread: annualized mean (%), volatility (%), CAPM alpha (%) against the market factor in Panel A and against the Nasdaq-100 benchmark (QQQ, in excess of the risk-free rate) in Panel B, its HC3 t -statistic, and the annualized Sharpe ratio. $n = 143$ monthly observations.

Panel A: Broad Universe										
	Excess return by score quintile					High-minus-low spread				
	Low	2	3	4	High	Mean	Vol	α	$t(\alpha)$	SR
<i>Constant</i>	1.4	7.6	9.0	10.0	11.1	9.8	11.5	11.2	3.41	0.85
<i>CAPM</i>	1.2	7.7	8.1	10.7	11.4	10.2	11.0	11.4	3.62	0.93
<i>FF3</i>	4.2	7.2	8.3	9.5	10.0	5.9	7.9	6.6	3.06	0.75
<i>FF6</i>	4.3	7.2	8.3	10.0	9.2	4.9	7.0	5.2	2.67	0.69
<i>JKP13</i>	4.2	7.0	9.1	9.4	9.4	5.2	6.2	5.1	2.81	0.83
<i>JKP</i>	5.2	7.4	8.6	8.6	9.3	4.0	4.3	4.0	3.12	0.95

Panel B: Nasdaq-100										
	Excess return by score quintile					High-minus-low spread				
	Low	2	3	4	High	Mean	Vol	α	$t(\alpha)$	SR
<i>Constant</i>	9.8	11.3	14.1	14.3	16.1	6.2	11.7	4.0	1.18	0.53
<i>CAPM</i>	9.6	13.6	11.2	16.9	14.2	4.6	11.7	2.6	0.73	0.39
<i>FF3</i>	12.7	8.7	16.1	13.5	14.4	1.7	11.0	0.4	0.12	0.16
<i>FF6</i>	11.2	10.6	12.9	16.5	14.2	3.0	10.7	1.3	0.40	0.28
<i>JKP13</i>	11.8	15.0	13.6	10.5	14.5	2.6	10.2	2.3	0.72	0.26

Table C.3: Pure News Portfolio Sorts Under Tile Spread Selection

This table shows equal-weighted quintile portfolios formed each month by sorting stocks on the predicted return of the pooled ridge, with the ridge penalty selected by the Tile Spread rule defined in the text, for each residualization specification. Panel A reports the broad universe, Panel B the Nasdaq-100. The first five columns report the annualized average return of each score quintile, in excess of the risk-free rate (%). The remaining columns report the high-minus-low spread: annualized mean (%), volatility (%), CAPM alpha (%) against the market factor in Panel A and against the Nasdaq-100 benchmark (QQQ, in excess of the risk-free rate) in Panel B, its HC3 t -statistic, and the annualized Sharpe ratio. $n = 143$ monthly observations.

Panel A: Broad Universe										
	Excess return by score quintile					High-minus-low spread				
	Low	2	3	4	High	Mean	Vol	α	$t(\alpha)$	SR
<i>Constant</i>	0.6	7.8	9.3	10.1	11.3	10.7	11.6	12.4	3.76	0.92
<i>CAPM</i>	0.2	7.8	8.5	11.2	11.5	11.3	11.1	12.8	4.02	1.02
<i>FF3</i>	3.4	7.7	8.0	9.5	10.5	7.1	8.2	8.0	3.43	0.87
<i>FF6</i>	3.9	7.3	8.0	10.4	9.6	5.8	7.3	6.2	3.08	0.79
<i>JKP13</i>	4.2	7.2	8.8	10.1	8.8	4.6	6.5	4.7	2.47	0.70
<i>JKP</i>	5.5	7.5	8.7	8.7	8.7	3.1	4.3	2.8	2.11	0.72

Panel B: Nasdaq-100										
	Excess return by score quintile					High-minus-low spread				
	Low	2	3	4	High	Mean	Vol	α	$t(\alpha)$	SR
<i>Constant</i>	9.7	12.0	10.5	16.0	17.3	7.6	11.2	5.5	1.72	0.68
<i>CAPM</i>	9.3	14.3	11.5	15.7	14.8	5.5	10.9	3.8	1.19	0.50
<i>FF3</i>	11.8	8.8	15.7	13.8	15.4	3.6	11.0	2.6	0.78	0.33
<i>FF6</i>	11.4	10.0	14.8	15.7	13.7	2.3	10.0	1.3	0.43	0.23
<i>JKP13</i>	11.7	16.3	12.7	11.4	13.4	1.7	9.8	1.6	0.54	0.17

Generative AI Disclaimer

We acknowledge the use of generative AI tools in the preparation of this thesis: OpenAI Codex, and Anthropic's Claude, used both through the Claude Code coding environment and as a chat assistant. Their use was not confined to specific chapters but transversal, in two capacities.

Text. French-to-English translation and reformulation of author-drafted text: rephrasing sentences for clarity and concision, correcting grammar, and proposing alternative formulations.

Code. Support in writing and debugging the analysis code, and in formatting figures.

Every AI-assisted passage was reviewed and validated by the author before inclusion.